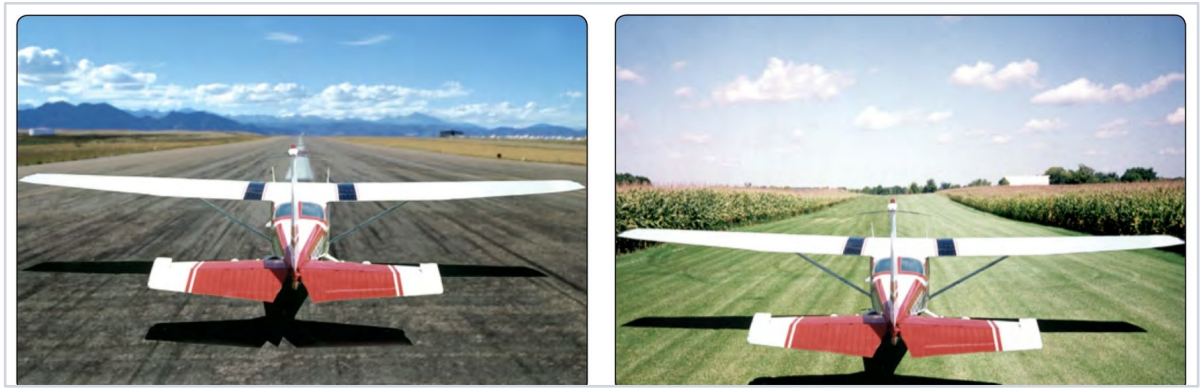
Performance and Limitations

Area II: Technical Subject Areas

ACS Task II.F (AI.II.F)

Cessna 172S NAV III (G1000)

[Name], CFI Applicant



Source: PHAK Fig 11-16

Objective

AI.II.F

- **Read POH performance charts**
- **Explain what changes performance**
- **Compute weight and CG**
- **Fix loading that is out of limits**

State the objective in plain words. The student will use the charts, tables, and data in the C172S POH to predict takeoff, climb, cruise, fuel, and landing performance. The student will compute weight and balance, correct a CG that is out of limits, and check that weight and balance stays in limits for the whole flight.

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

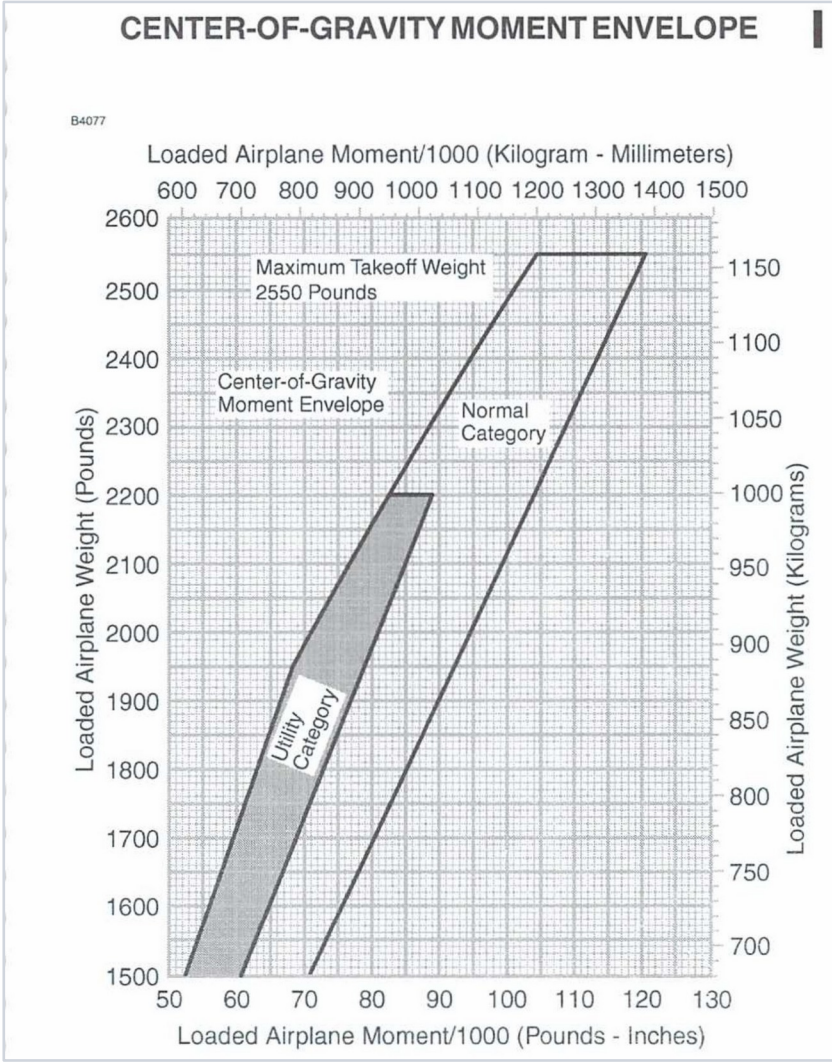
Source: ACS II.F, POH Fig 5-5

Lesson Overview

AI.II.F

- **Performance charts and factors**
- **Weight and balance terms**
- **C172S worked examples**
- **Risk management**

Give a short road map: charts and performance factors, weight and balance terms and methods, worked C172S examples, then risk management and review.



Source: POH Fig 6-7

AI.II.F.K1, K2

Performance Charts and Factors

What you will learn

- Where performance data lives
- How to read and interpolate charts
- Density altitude, technique, configuration
- Runway, wind, and weight

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

Why it matters: these charts tell you if the runway is long enough before you are rolling down it.

Source: PHAK 11-1 to 11-20, POH Section 5

Where the Numbers Live

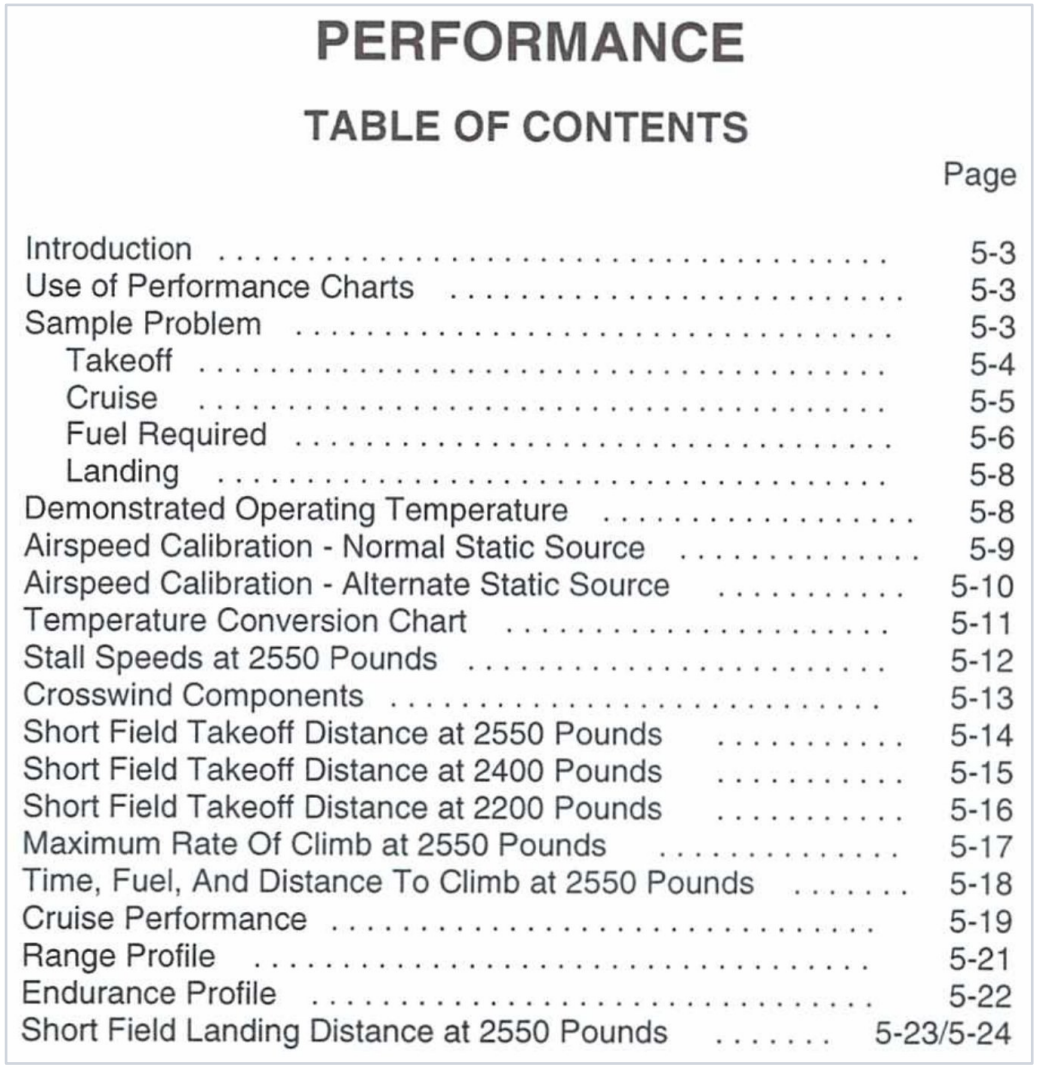
AI.II.F.K1

- **POH Section 2:** limitations
- **POH Section 5:** performance
- **POH Section 6:** weight and balance
- **TCDS:** certified limits

The performance section of the AFM/POH holds the data for takeoff, climb, range, endurance, descent, and landing. The PHAK says using this data is mandatory for safe and efficient operation (PHAK 11-1).

Show the C172S Section 5 table of contents. Point out each chart the student will use today: takeoff (Figure 5-5), climb (Figure 5-7), cruise (Figure 5-8), range (Figure 5-9), and landing (Figure 5-11).

CG limits are published in the Type Certificate Data Sheet (TCDS) and in the AFM or POH (PHAK 10-3).



PERFORMANCE

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Source: PHAK 11-1, PHAK 10-3, POH 5-1

How the Charts Are Made

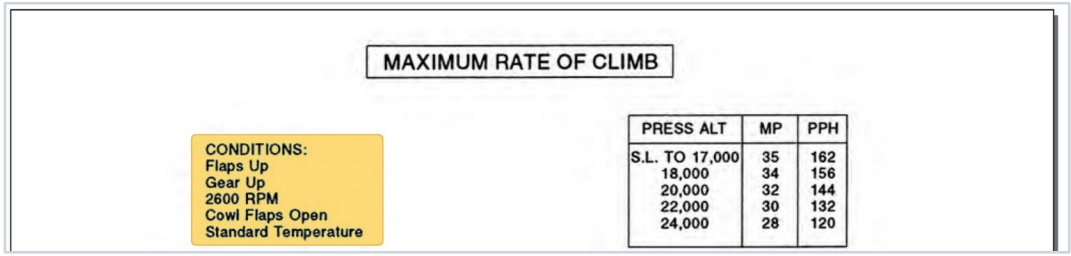
AI.II.F.K1

- **Flight tests in a new airplane**
- **Average piloting skill**
- **Airplane in good working order**
- **Always read the conditions**

Performance charts come from test flights conducted in a new aircraft, under normal operating conditions, using average piloting skills, with the aircraft and engine in good working order (PHAK 11-19).

The C172S POH says the data was computed from actual flight tests with the airplane and engine in good condition and approximating average piloting techniques (POH 5-3).

Every chart has a conditions block (flaps, power, runway, wind). If your situation does not match, the number does not match either.



MAXIMUM RATE OF CLIMB

CONDITIONS:
Flaps Up
Gear Up
2600 RPM
Cowl Flaps Open
Standard Temperature

PRESS ALT	MP	PPH
S.L. TO 17,000	35	162
18,000	34	156
20,000	32	144
22,000	30	132
24,000	28	120

KNOW THIS FOR THE CHECKRIDE

Chart numbers come from a new airplane flown with average skill (PHAK 11-19).

EXAMPLE

The C172S short field takeoff chart assumes flaps 10 degrees, full throttle before brake release, a paved, level, dry runway, and zero wind (POH Fig 5-5).

Source: PHAK 11-19, POH 5-3, PHAK Fig 11-20

Reading Tables and Interpolating

AI.II.F.K1

- **Find your row and column**
- **Interpolate between values**
- **Or use next higher value**
- **Higher value:** more margin

Interpolating means using known values on the chart to compute values in between (PHAK 11-20).

The PHAK example: for 2,500 feet pressure altitude at 20 degrees C, average the 2,000 foot and 3,000 foot ground rolls: (1,115 + 1,230) divided by 2 = 1,173 feet (PHAK Fig 11-21).

The C172S POH says conservative distances can be found by reading the chart at the next higher value of weight, altitude, and temperature (POH 5-4). This is faster and gives a safety margin.

Conditions	Flaps 10° Full throttle prior to brake release Paved level runway Zero wind			TAKEOFF DISTANCE MAXIMUM WEIGHT 2,400 LB									
Weight (lb)	Takeoff speed KIAS		Press ALT (ft)	0 °C		10 °C		20 °C		30 °C		40 °C	
	Lift off	AT 50 ft		Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS
2,400	51	56	S.L.	795	1,460	860	1,570	925	1,685	995	1,810	1,065	1,945
			1,000	875	1,605	940	1,725	1,015	1,860	1,090	2,000	1,170	2,155
			2,000	960	1,770	1,035	1,910	1,115	2,060	1,200	2,220	1,290	2,395
			3,000	1,055	1,960	1,140	2,120	1,230	2,295	1,325	2,480	1,425	2,685
			4,000	1,165	2,185	1,260	2,365	1,355	2,570	1,465	2,790	1,575	3,030
			5,000	1,285	2,445	1,390	2,660	1,500	2,895	1,620	3,160	1,745	3,455
			6,000	1,425	2,755	1,540	3,015	1,665	3,300	1,800	3,620	1,940	3,990
			7,000	1,580	3,140	1,710	3,450	1,850	3,805	2,000	4,220	- - -	- - -
			8,000	1,755	3,615	1,905	4,015	2,060	4,480	- - -	- - -	- - -	- - -

To find the takeoff distance for a pressure altitude of 2,500 feet at 20 °C, average the ground roll for 2,000 feet and 3,000 feet.

$$\frac{1,115 + 1,230}{2} = 1,173 \text{ feet}$$

KNOW THIS FOR THE CHECKRIDE

Not sure? Read the chart at the next higher weight, altitude, and temperature (POH 5-4).

Source: PHAK 11-20, POH 5-4, PHAK Fig 11-21

Definition: Density Altitude

AI.II.F.K2a

DENSITY ALTITUDE

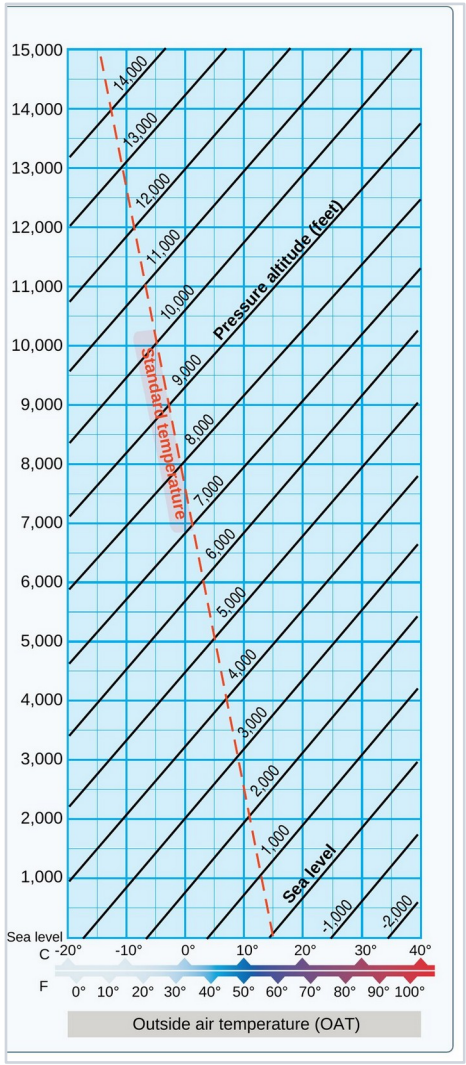
“the altitude in the standard atmosphere corresponding to a particular value of air density.”

ALSO STATED AS

“Density altitude is pressure altitude corrected for nonstandard temperature.”

IN PLAIN WORDS

Density altitude is the altitude the airplane feels like it is at. Hot, high, and humid air makes it feel higher, so it performs worse.



Source: PHAK 11-3, PHAK Fig 11-4

Density Altitude in Plain Words

AI.II.F.K2a

- **Hot, high, humid:** thin air
- **Less power, thrust, and lift**
- **Longer runs, weaker climbs**
- **Chart:** hotter column, bigger number

As air becomes less dense, it reduces power (the engine takes in less air), thrust (the propeller is less efficient in thin air), and lift (thin air exerts less force on the airfoils) (PHAK 11-2).

Moist air is lighter than dry air, so high humidity raises density altitude and lowers performance. There is no rule of thumb or chart for humidity, but expect a decrease in performance (PHAK 11-5).

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
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5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
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7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

MEMORY AID

Hot, high, and humid means poor performance.

EXAMPLE

On the C172S takeoff chart at 2,550 lb and sea level, total distance over a 50 foot obstacle grows from 1,465 feet at 0 degrees C to 1,945 feet at 40 degrees C. At 8,000 feet and 40 degrees C it is 4,615 feet (POH Fig 5-5).

Source: PHAK 11-2, PHAK 11-5, POH Fig 5-5

Pilot Technique

AI.II.F.K2b

- **Charts assume average technique**
- **Use the chart's exact speeds**
- **Short field technique:** Section 4
- **Weak technique:** add margin

The data assumes average piloting technique (POH 5-3). If skills are below average, the PHAK says to compensate for the chart numbers (PHAK 11-19).

The PHAK says proper takeoff speed is especially important when runway lengths and takeoff distances are critical (PHAK 11-15).

The C172S short field takeoff chart lists lift off at 51 KIAS and 56 KIAS at 50 feet, and the note says to use short field technique as specified in Section 4 (POH Fig 5-5).

- Prior to takeoff from fields above 3000 feet elevation, the mixture should be leaned to give maximum RPM in a full throttle, static runup.
- Decrease distances 10% for each 9 knots headwind. For operation with tail winds up to 10 knots, increase distances by 10% for each 2 knots.
- For operation on dry grass runway, increase distances by 15% of the "ground roll" figure.

Source: POH 5-3, POH Fig 5-5, PHAK 11-15, PHAK 11-19

Airplane Configuration

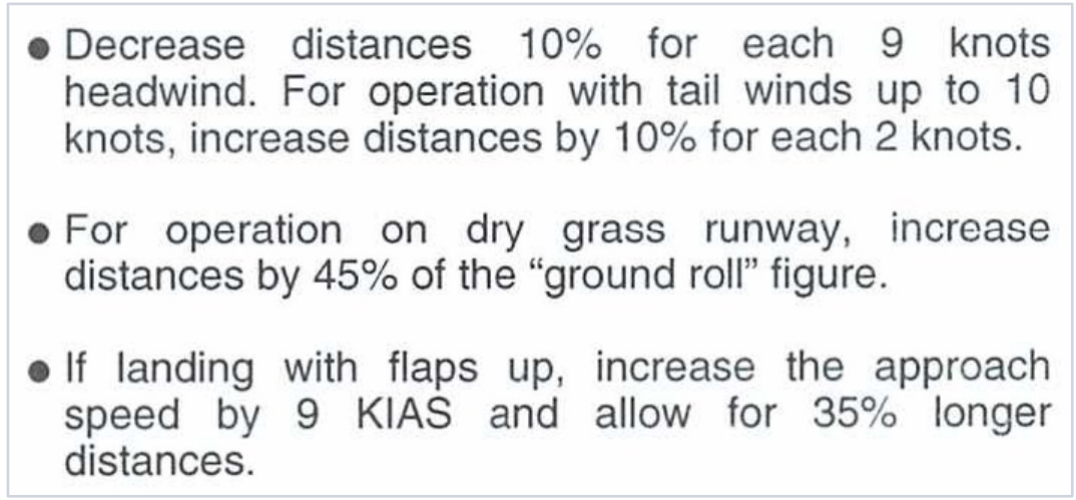
AI.II.F.K2c

- **Takeoff chart:** flaps 10 degrees
- **Climb chart:** flaps up
- **Flaps up landing:** 35% longer
- **Match the chart setup**

Configuration means flaps, gear, and power setting. Lowering the flaps decreases both excess thrust and excess power, which hurts climb (PHAK 11-8).

C172S examples: the short field takeoff chart assumes flaps 10 degrees (POH Fig 5-5). The climb chart assumes flaps up and full throttle (POH Fig 5-7). The short field landing chart assumes flaps full, power off, maximum braking (POH Fig 5-11).

Landing note: if landing with flaps up, increase the approach speed by 9 KIAS and allow for 35 percent longer distances (POH Fig 5-11).



- Decrease distances 10% for each 9 knots headwind. For operation with tail winds up to 10 knots, increase distances by 10% for each 2 knots.
- For operation on dry grass runway, increase distances by 45% of the "ground roll" figure.
- If landing with flaps up, increase the approach speed by 9 KIAS and allow for 35% longer distances.

KNOW THIS FOR THE CHECKRIDE

Flaps up landing in the C172S: add 9 KIAS and allow 35 percent more distance (POH Fig 5-11).

Source: POH Fig 5-5, POH Fig 5-7, POH Fig 5-11, PHAK 11-8

Airport Environment: Surface and Slope

AI.II.F.K2d

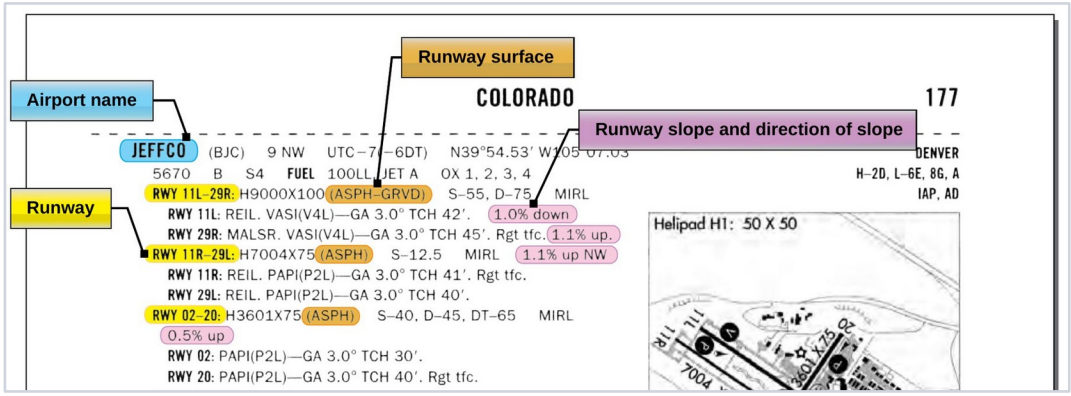
- **Charts assume paved, level, dry**
- **Grass takeoff:** +15% ground roll
- **Upslope:** longer takeoff roll
- **Check the Chart Supplement**

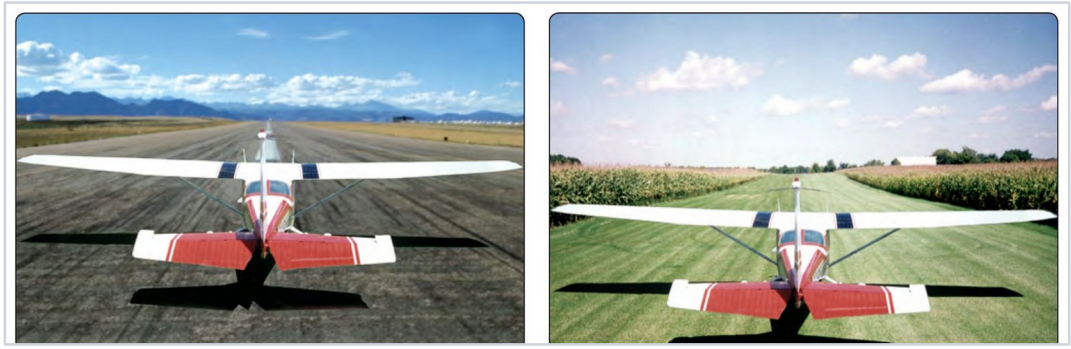
Performance charts typically assume paved, level, smooth, and dry runways. Any surface that is not hard and smooth increases the ground roll on takeoff (PHAK 11-12).

An upsloping runway gives a longer takeoff ground run but usually shortens the landing roll. A downslope does the opposite. Runway surface and slope are in the Chart Supplement (PHAK 11-13, PHAK Fig 11-17).

C172S notes: on a dry grass runway, increase takeoff distances by 15 percent of the ground roll figure (POH Fig 5-5) and landing distances by 45 percent of the ground roll figure (POH Fig 5-11).

Water on the runway reduces braking and can cause hydroplaning (PHAK 11-13).





Source: PHAK 11-12, PHAK 11-13, POH Fig 5-5, POH Fig 5-11, PHAK Fig 11-17

Airport Environment: Wind

AI.II.F.K2d

- **Headwind shortens distances**
- **Tailwind lengthens them more**
- **C172S:** minus 10% per 9 kt
- **Tailwind:** plus 10% per 2 kt

A headwind lets the airplane reach lift-off speed at a lower groundspeed. A tailwind requires a higher groundspeed (PHAK 11-15).

A headwind that is 10 percent of the takeoff airspeed reduces takeoff distance about 19 percent. A tailwind that is 10 percent of takeoff airspeed increases it about 21 percent (PHAK 11-15, PHAK Fig 11-19).

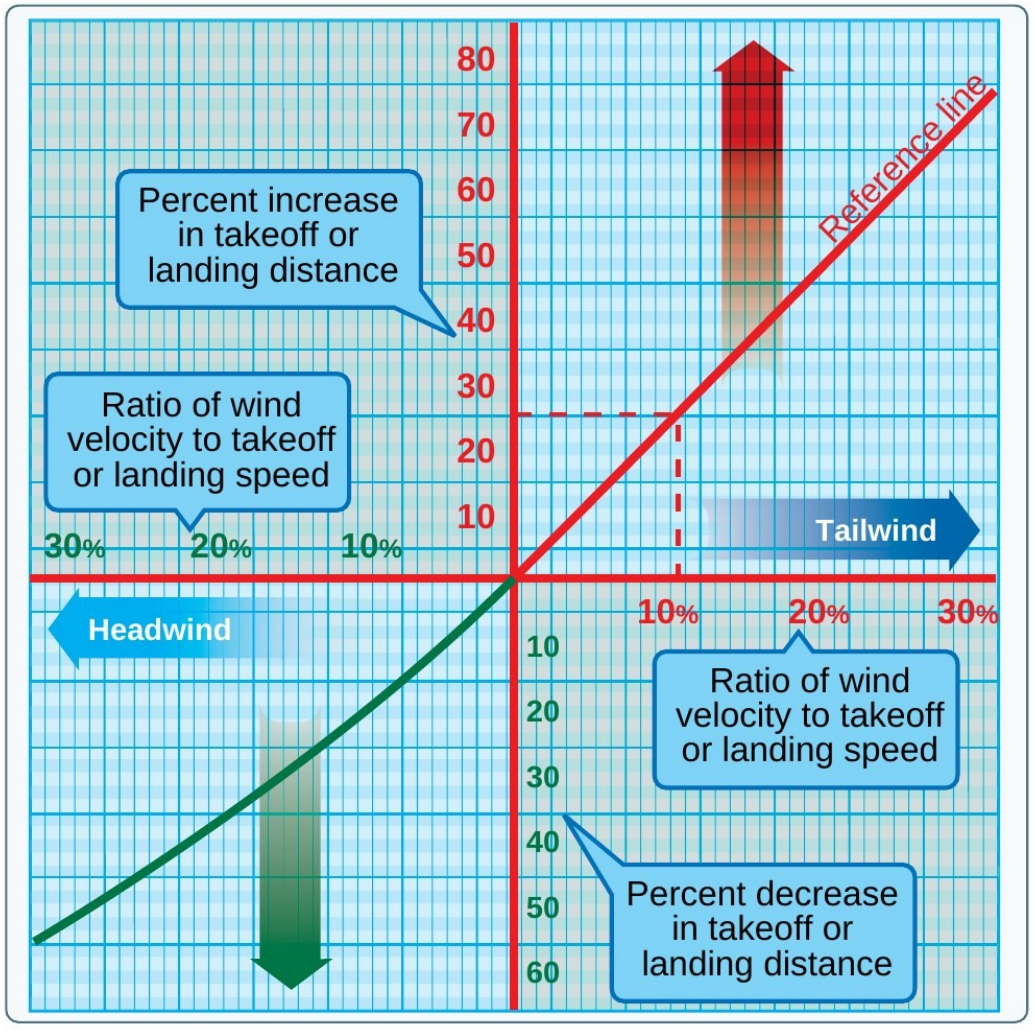
C172S note: decrease distances 10 percent for each 9 knots of headwind. For tailwinds up to 10 knots, increase distances by 10 percent for each 2 knots (POH Fig 5-5).

MEMORY AID

Headwind: minus 10 percent per 9 knots. Tailwind: plus 10 percent per 2 knots (POH Fig 5-5).

EXAMPLE

An 18 knot headwind: 18 divided by 9 = 2, times 10 percent = 20 percent shorter. A 4 knot tailwind: 4 divided by 2 = 2, times 10 percent = 20 percent longer.



Source: PHAK 11-15, POH Fig 5-5, PHAK Fig 11-19

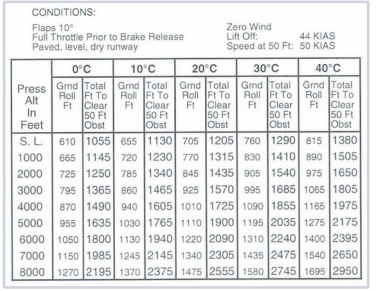
Loading and Weight

AI.II.F.K2e

- **Higher lift-off speed**
- **More mass to accelerate**
- **More drag and friction**
- **Lighter airplane:** shorter runs

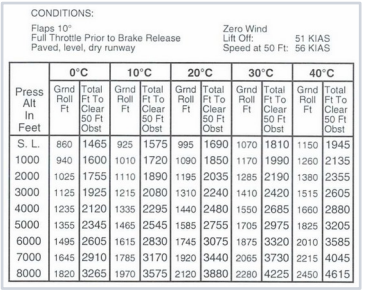
Increased gross weight has a threefold effect on takeoff performance: higher lift-off speed, greater mass to accelerate, and increased retarding force from drag and ground friction (PHAK 11-15).

Excess weight also gives a longer takeoff run, reduced climb, shorter range, higher stalling speed, higher approach speed, and a longer landing roll (PHAK 10-2).



CONDITIONS:
Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway
Zero Wind
Lift Off: 44 KIAS
Speed at 50 Ft: 50 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	610	1055	655	1130	705	1205	760	1290	815	1380
1000	665	1145	720	1230	770	1315	830	1410	890	1505
2000	725	1250	785	1340	845	1435	905	1540	975	1650
3000	795	1365	860	1465	925	1570	995	1685	1065	1805
4000	870	1490	940	1605	1010	1725	1090	1855	1165	1975
5000	955	1635	1030	1765	1110	1900	1195	2035	1275	2175
6000	1050	1800	1130	1940	1220	2090	1310	2240	1400	2395
7000	1150	1985	1245	2145	1340	2305	1435	2475	1540	2650
8000	1270	2195	1370	2375	1475	2555	1580	2745	1695	2950



CONDITIONS:
Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway
Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

KNOW THIS FOR THE CHECKRIDE

Extra weight means a higher lift-off speed, more mass to accelerate, and more drag and friction (PHAK 11-15).

EXAMPLE

C172S at sea level and 20 degrees C: total distance over a 50 foot obstacle is 1,690 feet at 2,550 lb (POH 5-14) and 1,205 feet at 2,200 lb (POH 5-16).

Source: PHAK 11-15, PHAK 10-2, POH 5-14, POH 5-16

Quick Check

AI.II.F.K1, K2

1 **What three things does thin air take away?**

Answer: Power, thrust, and lift (PHAK 11-2).

2 **Your takeoff distance is 2,000 feet with no wind. You have an 18 knot headwind. New distance?**

Answer: 18 / 9 x 10% = 20% shorter, so about 1,600 feet (POH Fig 5-5).

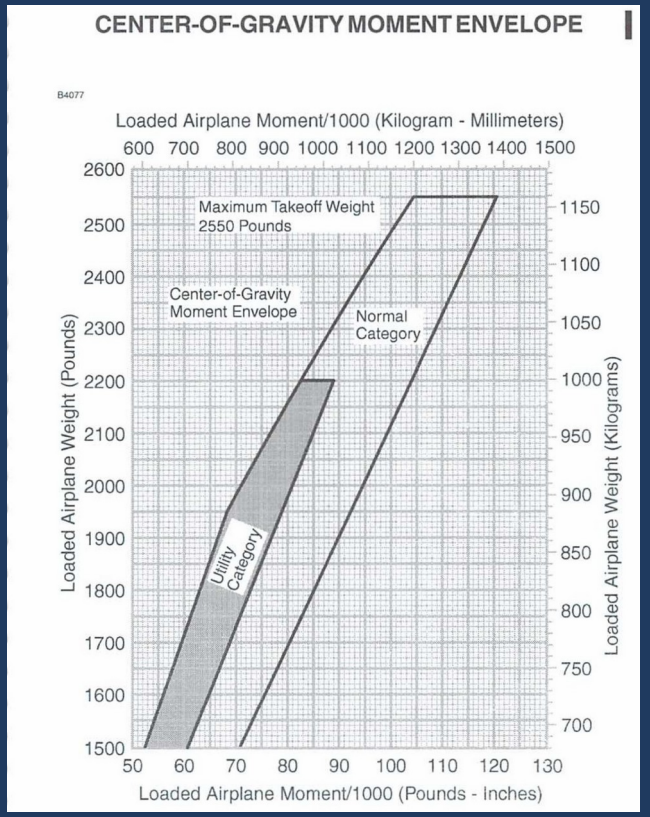
3 **Taking off from dry grass, what correction does the C172S chart give?**

Answer: Add 15 percent of the ground roll figure (POH Fig 5-5).

Source: PHAK 11-2, POH Fig 5-5

AI.II.F.K3 to K5

Weight and Balance



What you will learn

- The TCDS and C172S limits
- Weight and balance terms
- Three ways to compute CG
- Forward and aft CG effects

Why it matters: an airplane loaded out of limits can be hard or impossible to control.

Source: PHAK 10-1 to 10-11, POH 2-8, TCDS 3A12

The Type Certificate Data Sheet

AI.II.F.K3, AI.II.F.R2

- **Part of the type certificate**
- **Lists certified limits**
- **C172S:** TCDS 3A12
- **Engine, speeds, CG, weights**

TCDS 3A12 page 1 says the data sheet "prescribes conditions and limitations under which the product for which the type certificate was issued meets the airworthiness requirements."

The C172S section (page 30) lists the Lycoming IO-360-L2A rated 180 horsepower, 100/100LL minimum grade fuel, 2,700 RPM engine limit, and airspeed limits of 105 KIAS maneuvering, 129 KIAS maximum structural cruising, 163 KIAS never exceed, and 85 KIAS flaps extended.

CG limits are published in the TCDS and in the POH (PHAK 10-3). Use the POH for the airplane you fly, and know the TCDS backs it up.

XII. Model 172S, Skyhawk SP, 4 PCLM (Normal Category), 2 PCLM (Utility Category), Approved May 1, 1998

Engine	Lycoming IO-360-L2A, Rated 180 Horsepower
Fuel	100/100LL minimum grade aviation gasoline
Engine Limits	For all operations, 2,700 RPM
Propeller and Propeller Limits	1. McCauley Propeller Hub 1A170E/JHA7660

C.G. Range	Normal Category	
	(1) Aft Limits	47.3 inches aft of datum at 2,550 pounds or less.
	(2) Forward Limits	Linear variation from 41.0 inches aft of datum at 2,550 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
	Utility Category	
	(1) Aft Limits	40.5 inches aft of datum at 2,200 pounds or less.
	(2) Forward Limits	Linear variation from 37.5 inches aft of datum at 2,200 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
Empty Wt. C.G. Range	None	
Reference Datum	Lower portion of front face of firewall	
MAC	58.8 inches; Leading edge of MAC 25.9 inches aft of datum	
Leveling Means	Left side of Tailcone at 108.0 inches and 142.0 inches aft of datum	

Source: TCDS 3A12 p1, TCDS 3A12 p30, PHAK 10-3

Definitions: Datum and Arm

AI.II.F.K3

DATUM (REFERENCE DATUM)

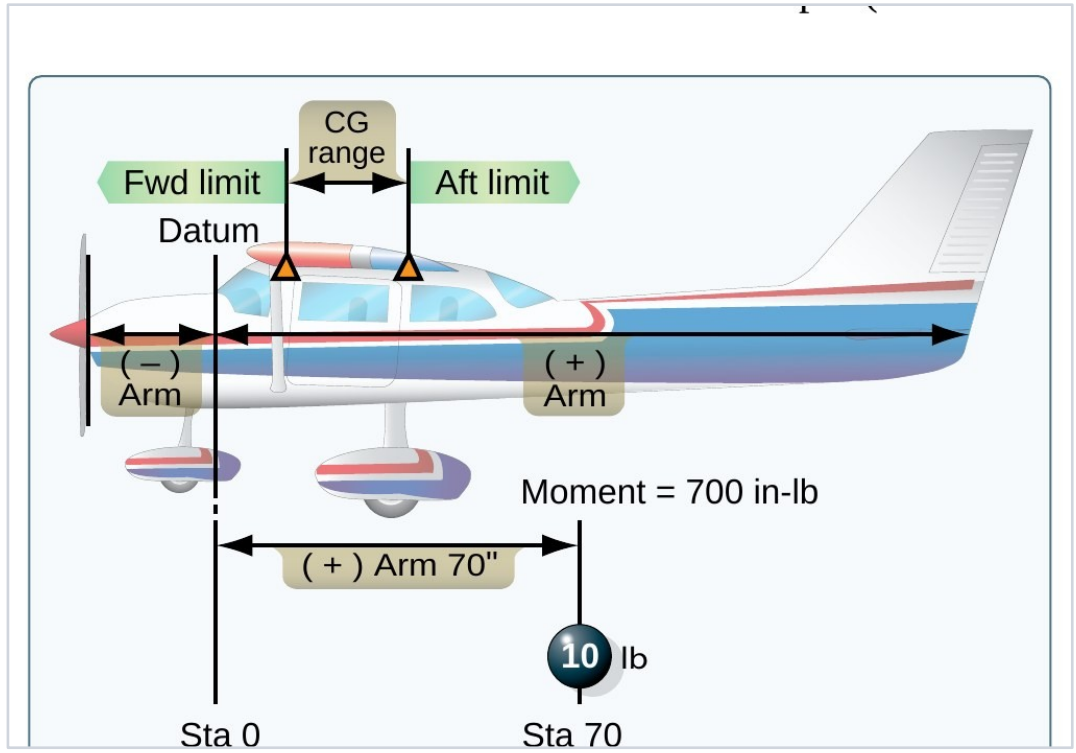
“an imaginary vertical plane or line from which all measurements of arm are taken. The datum is established by the manufacturer. Once the datum has been selected, all moment arms and the location of CG range are measured from this point.”

ARM (MOMENT ARM)

“the horizontal distance in inches from the reference datum line to the CG of an item. The algebraic sign is plus (+) if measured aft of the datum and minus (−) if measured forward of the datum.”

IN PLAIN WORDS

The datum is the zero point on a tape measure. The arm is how far each seat, tank, or bag sits from that zero point.



Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-2

Datum and Arm in Plain Words

AI.II.F.K3

- **Datum:** the zero point
- **C172S:** front face of firewall
- **Arm:** inches from datum
- **Aft is plus, forward minus**

The datum is the zero point for all measurements. For the C172S it is the lower portion of the front face of the firewall (TCDS 3A12 p30).

The arm is how far an item is from the datum. Items behind the datum are plus, items in front are minus (PHAK 10-6).

C.G. Range	Normal Category	
	(1) Aft Limits	47.3 inches aft of datum at 2,550 pounds or less.
	(2) Forward Limits	Linear variation from 41.0 inches aft of datum at 2,550 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
	Utility Category	
	(1) Aft Limits	40.5 inches aft of datum at 2,200 pounds or less.
	(2) Forward Limits	Linear variation from 37.5 inches aft of datum at 2,200 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
Empty Wt. C.G. Range	None	
Reference Datum	Lower portion of front face of firewall	
MAC	58.8 inches; Leading edge of MAC 25.9 inches aft of datum	
Leveling Means	Left side of Tailcone at 108.0 inches and 142.0 inches aft of datum	

EXAMPLE

C172S arms from the TCDS: front seats 34.0 to 46.0 inches, rear seats 73.0 inches, fuel 48.0 inches aft of datum, and oil 13.1 inches forward of datum, which is a minus arm (TCDS 3A12 p30).

THINK OF IT LIKE THIS

hook a tape measure on the firewall and stretch it back. The number on the tape under each seat or tank is its arm.

Source: TCDS 3A12 p30, PHAK 10-6

Definitions: Moment and CG

AI.II.F.K3

MOMENT

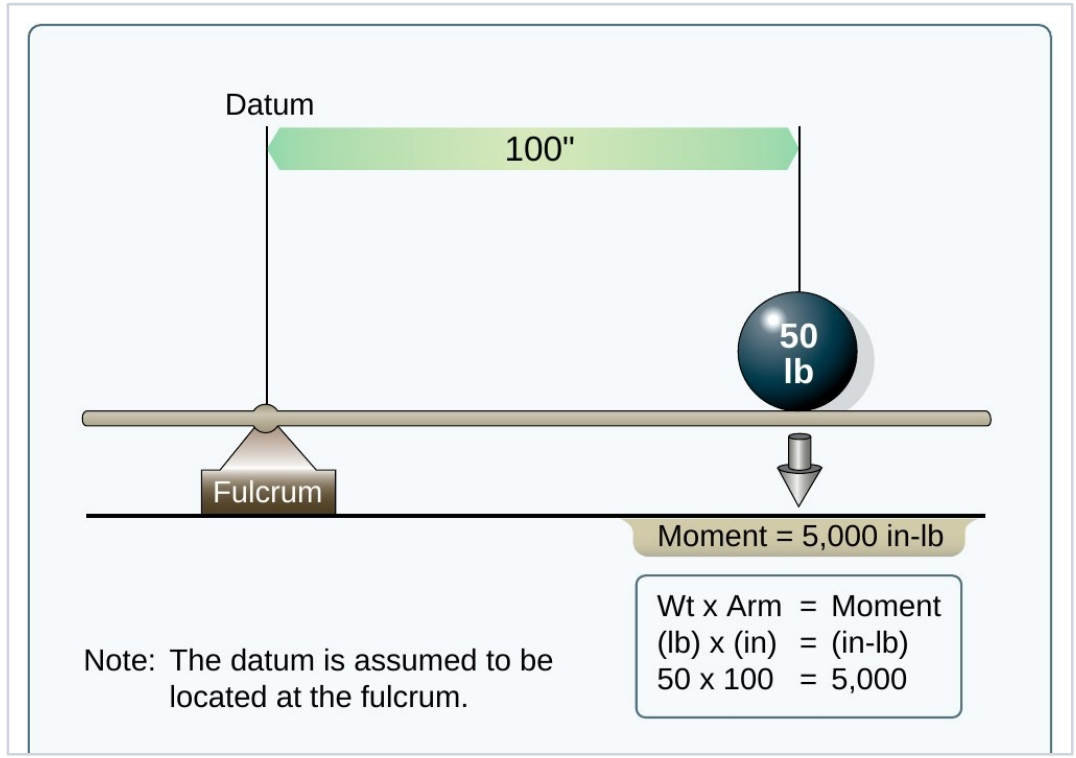
"the product of the weight of an item multiplied by its arm. Moments are expressed in pound-inches (in-lb). Total moment is the weight of the airplane multiplied by the distance between the datum and the CG."

CENTER OF GRAVITY (CG)

"the point about which an aircraft would balance if it were possible to suspend it at that point."

IN PLAIN WORDS

Moment is how hard a weight tries to tip the airplane. The CG is the balance point of everything combined.



Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-3

Moment and CG in Plain Words

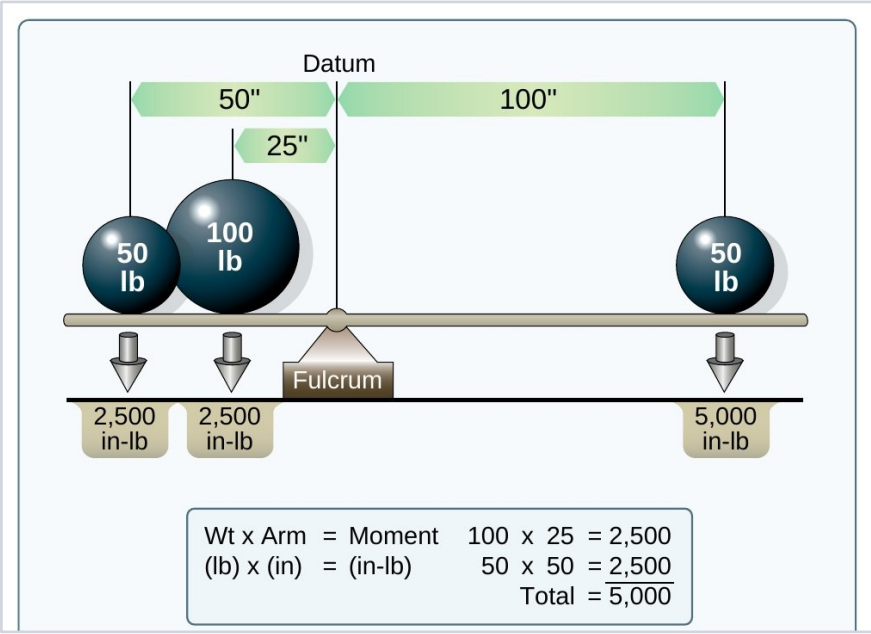
AI.II.F.K3

- **Moment = weight x arm**
- **50 lb x 100 in = 5,000**
- **CG:** the balance point
- **CG = total moment / weight**

A moment is weight times arm. It measures how strongly a weight tries to rotate about a point (PHAK 10-6).

PHAK example: 50 lb placed 100 inches from the datum makes 5,000 in-lb. That same 5,000 in-lb can be balanced by 100 lb at 25 inches plus 50 lb at 50 inches (PHAK Fig 10-4).

To find the CG, divide total moment by total weight (PHAK 10-7).



MEMORY AID

Moment = Weight x Arm. CG = Total Moment / Total Weight.

THINK OF IT LIKE THIS

a playground seesaw. A light child far from the middle can balance a heavy child close to the middle. The seesaw balances where the moments are equal.

Source: PHAK 10-6, PHAK 10-7, PHAK Fig 10-4

Definitions: CG Limits and Maximum Weight

AI.II.F.K3

CG LIMITS

“the specified forward and aft points within which the CG must be located during flight. These limits are indicated on pertinent aircraft specifications.”

MAXIMUM WEIGHT

“the maximum authorized weight of the aircraft and all of its equipment as specified in the TCDS for the aircraft.”

IN PLAIN WORDS

CG limits are the front and back edges of the safe balance zone. Maximum weight is the heaviest the airplane may be.



CENTER-OF-GRAVITY LIMITS

NORMAL CATEGORY

Center-of-Gravity Range:

Forward:	35.0 inches aft of datum at 1950 pounds or less, with straight line variation to 41.0 inches aft of datum at 2550 pounds.
Aft:	47.3 inches aft of datum at all weights.

Reference Datum: Lower portion of front face of firewall.

Source: PHAK 10-5, POH 2-8

C172S Weight and CG Limits

AI.II.F.K3, AI.II.F.R2

- **Max takeoff and landing:** 2,550 lb
- **Max ramp:** 2,558 lb
- **Forward:** 35.0 to 41.0 in
- **Aft:** 47.3 in, all weights

Normal category limits: maximum ramp weight 2,558 lb, maximum takeoff and landing weight 2,550 lb. Baggage area A (station 82 to 108) holds 120 lb, baggage area B (station 108 to 142) holds 50 lb, and the combined maximum is 120 lb (POH 2-8).

CG range: forward limit 35.0 inches aft of datum at 1,950 lb or less, with straight line variation to 41.0 inches at 2,550 lb. Aft limit 47.3 inches at all weights (POH 2-8, TCDS 3A12 p30).

Utility category has smaller limits: 2,200 lb maximum takeoff weight, aft limit 40.5 inches, and the baggage area must be empty with the rear seat unoccupied (POH 2-8, POH 2-9).



WEIGHT LIMITS

NORMAL CATEGORY

Maximum Ramp Weight:	2558 POUNDS
Maximum Takeoff Weight:	2550 POUNDS
Maximum Landing Weight:	2550 POUNDS

Maximum Weight in Baggage Compartment:

Baggage Area A - Station 82 to 108:	120 POUNDS
Baggage Area B - Station 108 to 142:	50 POUNDS

KNOW THIS FOR THE CHECKRIDE

Max takeoff 2,550 lb. CG range 35.0 to 47.3 inches aft of the firewall (POH 2-8).

Source: POH 2-8, TCDS 3A12 p30

Definitions: Basic Empty Weight and Useful Load

AI.II.F.K3

BASIC EMPTY WEIGHT (GAMA)

"the standard empty weight plus the weight of optional and special equipment that have been installed."

USEFUL LOAD

"the weight of the pilot, copilot, passengers, baggage, usable fuel, and drainable oil. It is the basic empty weight subtracted from the maximum allowable gross weight."

IN PLAIN WORDS

Basic empty weight is the airplane with unusable fuel and full oil, nothing else. Useful load is what is left for people, bags, and fuel.

SAMPLE LOADING PROBLEM

ITEM DESCRIPTION	WEIGHT AND MOMENT TABULATION			
	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (Lbs.)	Moment (Lb-ins. /1000)	Weight (Lbs.)	Moment (Lb-ins. /1000)
1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil)	1642	62.6		
2. Usable Fuel (At 6 Lbs./Gal.)				
Standard Fuel 53 Gallons Maximum				
Reduced Fuel (30 Gallons)	180	8.6		
3. Pilot and Front Passenger (FS 32 to 50)	340	12.6		
4. Rear Passengers (FS 74)	340	24.8		
5.* Baggage "A" (FS 82 to 108) 120 Pounds Maximum	56	4.6		
6.* Baggage "B" (FS 108 to 142) 50 Pounds Maximum				
7. **RAMP WEIGHT AND MOMENT**	2558	113.2		
8. Fuel allowance for engine start, taxi and runup	-8.0	-0.4		
9. **TAKEOFF WEIGHT AND MOMENT** (Subtract Step 8 from Step 7)	2550	112.8		

10. Locate this point (2550 at 112.8) on the Center-of-Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

* The maximum allowable combined weight capacity for baggage in areas A and B is 120 pounds.

Source: PHAK 10-4, PHAK 10-5, POH Fig 6-3

Empty Weight and Useful Load in Plain Words

AI.II.F.K3

- **BEW:** your airplane's records
- **Includes unusable fuel, full oil**
- **Useful load:** max minus BEW
- **Sample:** 2,550 minus 1,642 = 908

Basic empty weight comes from the weight and balance records for your airplane as it is presently equipped. It includes unusable fuel and full oil (POH 6-10). TCDS Note 1 says the empty weight must include 18 lb of unusable fuel at 46.0 inches and 15.0 lb of full oil at 13.1 inches forward of datum (TCDS 3A12 p32).

Useful load is everything we can add: people, bags, and usable fuel.

Maximum Weights	Normal Category	
	Maximum Ramp	2,558 lb.
	Maximum Takeoff and Landing	2,550 lb.
	Utility Category	
	Maximum Ramp	2,208 lb.
	Maximum Takeoff and Landing	2,200 lb.
No. of Seats	4 (2 at 34.0 to 46.0 inches aft of datum; 2 at 73.0 inches aft of datum)	
Maximum Baggage	120 lb. at 82.0 to 108.0 inches aft of datum 50 lb. at 108.0 to 142.0 inches aft of datum (Max. combined weight capacity for baggage areas is 120 pounds)	
Fuel Capacity (Gal.)	56 gallons total; 53 gallons usable (two 28-gallon tanks in wings at 48.0 inches aft of datum). (See NOTE 1 for data on unusable fuel).	
Oil Capacity (Gal.)	8.0 quarts at 13.1 inches forward of datum 3.0 quarts usable	

EXAMPLE

The POH sample airplane has a basic empty weight of 1,642 lb (POH 6-10). 2,550 lb maximum takeoff weight minus 1,642 lb = 908 lb useful load. The C172S holds 53 gallons usable fuel (TCDS 3A12 p30). At 6 lb per gallon that is 318 lb, leaving 590 lb for people and bags.

THINK OF IT LIKE THIS

a pickup truck with a payload rating. The truck itself is the empty weight. What you can put in the bed and the cab is the useful load.

Source: POH 6-10, TCDS 3A12 p30, TCDS 3A12 p32, PHAK 10-6

Three Ways to Compute CG

AI.II.F.K4

- **Computation:** weight x arm
- **Graph:** loading graph, envelope
- **Table:** moments already figured
- **Same answer every way**

Computational method: list each weight, multiply by its arm to get the moment, total the weights and moments, then divide total moment by total weight to get the CG (PHAK 10-7).

PHAK example: total weight 3,320 lb, total moment 281,430 in-lb. 281,430 divided by 3,320 = 84.8 inches, inside the 78 to 86 inch range (PHAK Fig 10-5).

Graph method: use the manufacturer's loading graph to find moments, then plot total weight and moment on the CG envelope (PHAK 10-7). Table method: the manufacturer gives tables of moments for each weight (PHAK 10-9).

Item	Weight	Arm	Moment
Aircraft Empty Weight	2,100	78.3	164,430
Front Seat Occupants	340	85.0	28,900
Rear Seat Occupants	350	121.0	42,350
Fuel	450	75.0	33,750
Baggage Area 1	80	150.0	12,000
Total	3,320		281,430
			281,430 ÷ 3,320 = 84.8

Source: PHAK 10-7, PHAK 10-9, PHAK Fig 10-5

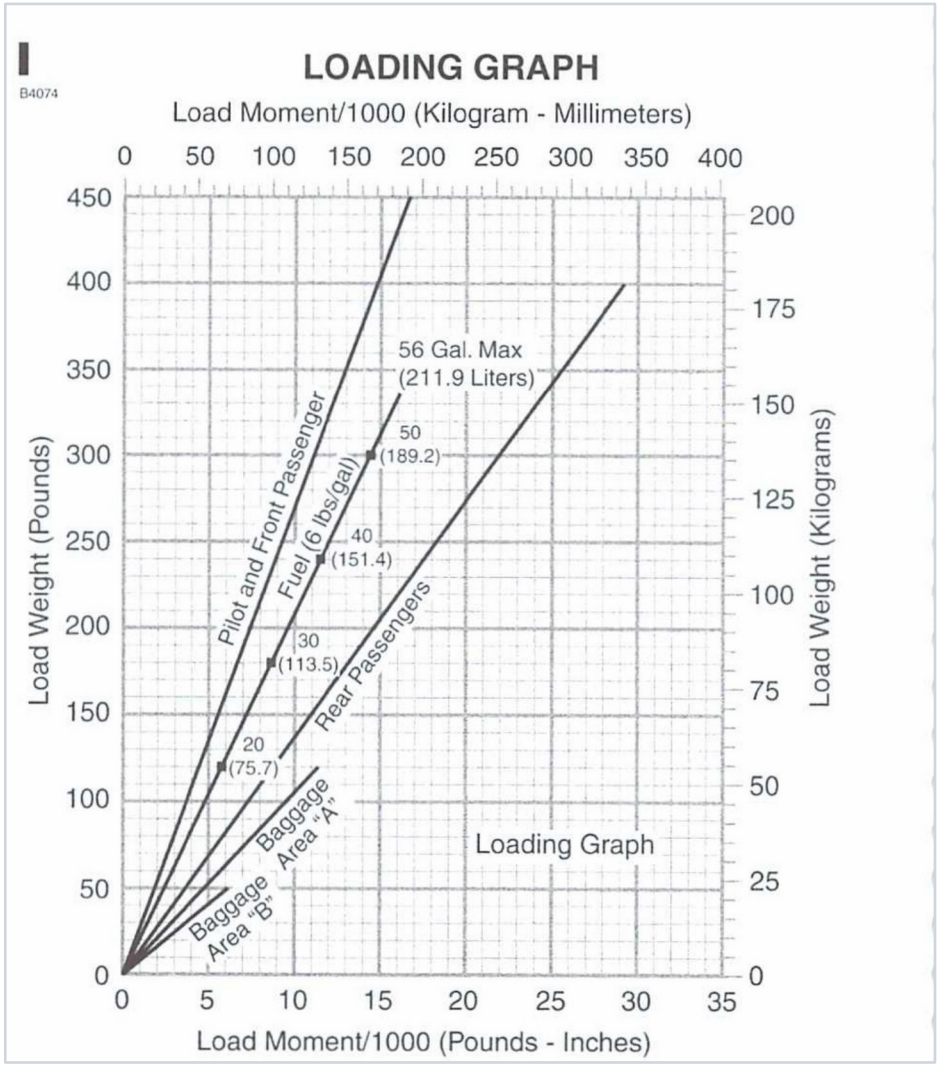
C172S Graph Method

AI.II.F.K4

- **Find load weight on left**
- **Go across to the item line**
- **Drop down:** moment / 1000
- **Add up, then plot envelope**

Using the loading graph: find the weight, draw a line straight across until it meets the line for the item, then draw a line straight down to read the moment (PHAK 10-7).

The C172S loading graph has lines for pilot and front passenger, fuel, rear passengers, baggage area A, and baggage area B. The moment scale is load moment divided by 1,000 in pound-inches (POH Fig 6-4).



Source: PHAK 10-7, POH Fig 6-4

Aerodynamics: Forward CG

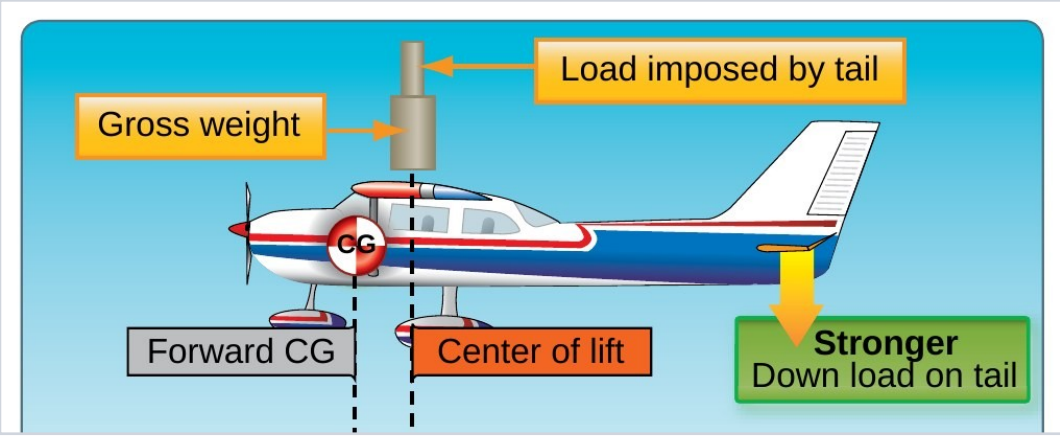
AI.II.F.K5

- **Tail pushes down harder**
- **Higher stall speed**
- **Heavier elevator forces**
- **Harder to flare**

With forward loading, the tail must produce a greater down load. That adds to the wing loading, requires a higher angle of attack, adds drag, and raises the stall speed (PHAK 5-43).

The airplane stalls at a higher speed with a forward CG, and elevator control forces are higher (PHAK 5-44).

Beyond the forward limit the nose may be so heavy that it is difficult or impossible to flare for landing (PHAK 10-3).



KNOW THIS FOR THE CHECKRIDE

Forward CG: higher stall speed and heavier control forces (PHAK 5-44).

Source: PHAK 5-43, PHAK 5-44, PHAK 10-3

Aerodynamics: Aft CG

AI.II.F.K5

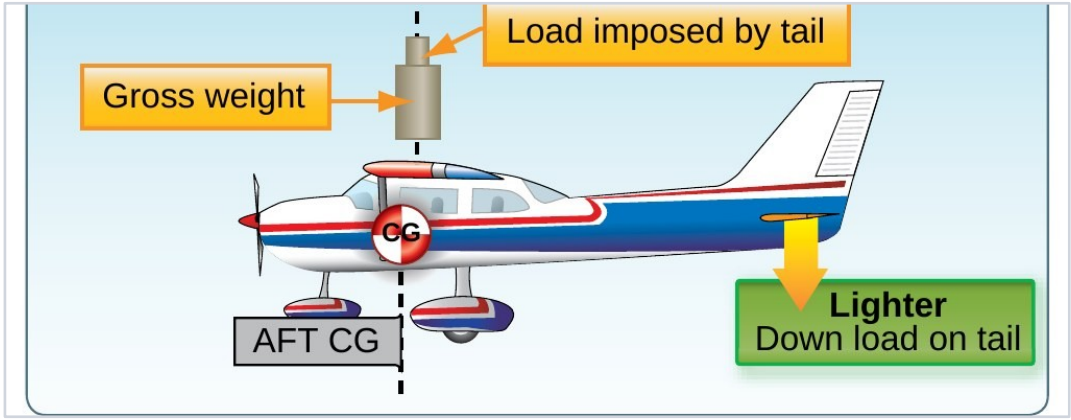
- **Less stable**
- **Very light control forces**
- **Harder stall and spin recovery**
- **Faster cruise from less drag**

The airplane becomes less stable as the CG moves aft. If it moves far enough aft the airplane becomes unstable (PHAK 5-44).

Aft of the limit: reduced stability, extreme control difficulty, violent stall characteristics, and very light control forces that make it easy to overstress the airplane (PHAK 10-3).

Stall recovery gets harder as the CG moves aft, and a flat spin can develop that may be impossible to recover from (PHAK 5-43, PHAK 5-44).

An aft CG cruises faster because of reduced drag (PHAK 5-44). This is the trap: it feels better until you need control at low speed.



KNOW THIS FOR THE CHECKRIDE

Aft CG: less stable, light controls, harder stall and spin recovery (PHAK 10-3).

Source: PHAK 5-43, PHAK 5-44, PHAK 10-3

Quick Check

AI.II.F.K3 to K5

1 **How do you find the CG?**

Answer: Total moment divided by total weight (PHAK 10-7).

2 **What is the C172S forward CG limit at 2,550 lb?**

Answer: 41.0 inches aft of datum (POH 2-8).

3 **Does a forward CG raise or lower the stall speed?**

Answer: Raises it (PHAK 5-44).

Source: PHAK 10-7, POH 2-8, PHAK 5-44

AI.II.F.S1, S2

Worked C172S Examples

What you will learn

- Takeoff distance with wind
- Climb fuel and cruise
- Total fuel required
- Loading, CG, and fuel burn

CRUISE PERFORMANCE

CONDITIONS:
2550 Pounds
Recommended Lean Mixture At All Altitudes (Refer to Section 4, Cruise)

PRESS ALT FT	RPM	20°C BELOW STANDARD TEMP			STANDARD TEMPERATURE			20°C ABOVE STANDARD TEMP		
		% MCP	KTAS	GPH	% MCP	KTAS	GPH	% MCP	KTAS	GPH
2000	2550	83	117	11.1	77	118	10.5	72	117	9.9
	2500	78	115	10.6	73	115	9.9	68	115	9.4
	2400	69	111	9.6	64	110	9.0	60	109	8.5
	2300	61	105	8.6	57	104	8.1	53	102	7.7
	2200	53	99	7.7	50	97	7.3	47	95	6.9
	2100	47	92	6.9	44	90	6.6	42	89	6.3
4000	2600	83	120	11.1	77	120	10.4	72	119	9.8
	2550	79	118	10.6	73	117	9.9	68	117	9.4
	2500	74	115	10.1	69	115	9.5	64	114	8.9
	2400	65	110	9.1	61	109	8.5	57	107	8.1
	2300	58	104	8.2	54	102	7.7	51	101	7.3
	2200	51	98	7.4	48	96	7.0	45	94	6.7
	2100	45	91	6.6	42	89	6.4	40	87	6.1
6000	2650	83	122	11.1	77	122	10.4	72	121	9.8
	2600	78	120	10.6	73	119	9.9	68	118	9.4
	2500	70	115	9.6	65	114	9.0	60	112	8.5
	2400	62	109	8.6	57	108	8.2	54	106	7.7
	2300	54	103	7.8	51	101	7.4	48	99	7.0
	2200	48	96	7.1	45	94	6.7	43	92	6.4

Why it matters: this is exactly what you will do with a student before a cross-country.

Source: POH 5-3 to 5-8, POH 6-10

Example: C172S Takeoff Setup

AI.II.F.S1, AI.II.F.K1

- **POH sample problem**
- **Read next higher values**

Item	Given
Takeoff weight	2,550 lb
Field pressure altitude	1,500 ft
Temperature	28 C
Wind	12 kt headwind
Field length	3,500 ft
Chart row used	2,000 ft, 30 C

Use the sample problem from POH 5-3 and 5-4. Have the student find the 2,000 foot row and the 30 degree C column on Figure 5-5.

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

Source: POH 5-3, POH 5-4, POH Fig 5-5

Example: Takeoff with Wind Correction

AI.II.F.S1

- **12/9 x 10% = 13% shorter**

Distance (ft)	Chart	Minus 13%	Corrected
Ground roll	1,285	167	1,118
Over 50 ft obstacle	2,190	285	1,905

From Figure 5-5 at 2,000 feet and 30 degrees C: ground roll 1,285 feet, total to clear a 50 foot obstacle 2,190 feet (POH 5-4).

Wind correction from Note 3: 12 knots divided by 9 knots, times 10 percent, is about a 13 percent decrease (POH 5-4).

Corrected ground roll: 1,285 minus 167 = 1,118 feet. Corrected total distance: 2,190 minus 285 = 1,905 feet (POH 5-5). Both fit in the 3,500 foot field.

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

Source: POH 5-4, POH 5-5, POH Fig 5-5

Example: Fuel and Distance to Climb

AI.II.F.S1

- **2,000 to 8,000 ft:** 2.2 gal
- **Hot day:** +16% (16 C)
- **Corrected fuel:** 2.6 gal
- **Distance:** 18 nm

Figure 5-7 shows a climb from 2,000 to 8,000 feet takes 2.2 gallons and 15 nautical miles at standard temperature (POH 5-6).

The chart note says increase time, fuel, and distance by 10 percent for each 10 degrees C above standard. At 16 degrees above standard that is a 16 percent increase (POH 5-6, POH Fig 5-7).

Fuel: 2.2 plus 0.4 = 2.6 gallons. Distance: 15 plus 2.4 = 17.4, rounded up to 18 nautical miles (POH 5-6).

EXAMPLE

Show how to subtract: at 8,000 feet the chart reads 3.0 gallons and 19 nm from sea level. At 2,000 feet it reads 0.8 gallons and 4 nm. The POH sample uses 2.2 gallons and 15 nm for the climb between them (POH Fig 5-7).

TIME, FUEL AND DISTANCE TO CLIMB AT 2550 POUNDS

CONDITIONS:

Flaps Up
Full Throttle
Standard Temperature

PRESS ALT FT	CLIMB SPEED KIAS	RATE OF CLIMB FPM	FROM SEA LEVEL		
			TIME IN MIN	FUEL USED GAL	DIST NM
S.L.	74	730	0	0.0	0
1000	73	695	1	0.4	2
2000	73	655	3	0.8	4
3000	73	620	4	1.2	6
4000	73	600	6	1.5	8
5000	73	550	8	1.9	10
6000	73	505	10	2.2	13
7000	73	455	12	2.6	16
8000	72	410	14	3.0	19
9000	72	360	17	3.4	22
10,000	72	315	20	3.9	27
11,000	72	265	24	4.4	32
12,000	72	220	28	5.0	38

NOTE

- Add 1.4 gallons of fuel for engine start, taxi and takeoff allowance.
- Mixture leaned above 3000 feet elevation for maximum RPM.
- Increase time, fuel and distance by 10% for each 10°C above standard temperature.
- Distances shown are based on zero wind.

Source: POH 5-6, POH Fig 5-7

Example: Cruise Performance

AI.II.F.S1

- **8,000 ft, 20 C above standard**
- **2,600 RPM:** 64% power
- **117 KTAS at 8.9 GPH**
- **10 kt headwind:** 107 kt GS

Cruise is planned at 7,500 feet pressure altitude, 16 degrees C. The cruise chart is entered at 8,000 feet and 20 degrees C above standard, which most nearly match (POH 5-5).

At 2,600 RPM: 64 percent power, 117 knots true airspeed, 8.9 gallons per hour (POH 5-5, POH Fig 5-8).

With a 10 knot headwind, groundspeed is 117 minus 10 = 107 knots (POH 5-7).

The range profile shows lower power settings save fuel and give longer range (POH 5-5).

CRUISE PERFORMANCE

CONDITIONS:
2550 Pounds
Recommended Lean Mixture At All Altitudes (Refer to Section 4, Cruise)

PRESS ALT FT	RPM	20°C BELOW STANDARD TEMP			STANDARD TEMPERATURE			20°C ABOVE STANDARD TEMP		
		% MCP	KTAS	GPH	% MCP	KTAS	GPH	% MCP	KTAS	GPH
2000	2550	83	117	11.1	77	118	10.5	72	117	9.9
	2500	78	115	10.6	73	115	9.9	68	115	9.4
	2400	69	111	9.6	64	110	9.0	60	109	8.5
	2300	61	105	8.6	57	104	8.1	53	102	7.7
	2200	53	99	7.7	50	97	7.3	47	95	6.9
	2100	47	92	6.9	44	90	6.6	42	89	6.3
4000	2600	83	120	11.1	77	120	10.4	72	119	9.8
	2550	79	118	10.6	73	117	9.9	68	117	9.4
	2500	74	115	10.1	69	115	9.5	64	114	8.9
	2400	65	110	9.1	61	109	8.5	57	107	8.1
	2300	58	104	8.2	54	102	7.7	51	101	7.3
	2200	51	98	7.4	48	96	7.0	45	94	6.7
	2100	45	91	6.6	42	89	6.4	40	87	6.1
6000	2650	83	122	11.1	77	122	10.4	72	121	9.8
	2600	78	120	10.6	73	119	9.9	68	118	9.4
	2500	70	115	9.6	65	114	9.0	60	112	8.5
	2400	62	109	8.6	57	108	8.2	54	106	7.7
	2300	54	103	7.8	51	101	7.4	48	99	7.0
	2200	48	96	7.1	45	94	6.7	43	92	6.4

Source: POH 5-5, POH 5-7, POH Fig 5-8

Example: Total Fuel Required

AI.II.F.S1

- **360 nm trip, 53 gal usable**

Leg	Gallons
Start, taxi, takeoff	1.4
Climb (18 nm)	2.6
Cruise (342 nm, 3.2 hr)	28.5
Reserve (45 min)	6.7
Total required	**39.2**

Cruise distance: 360 minus 18 climb miles = 342 nautical miles (POH 5-6).

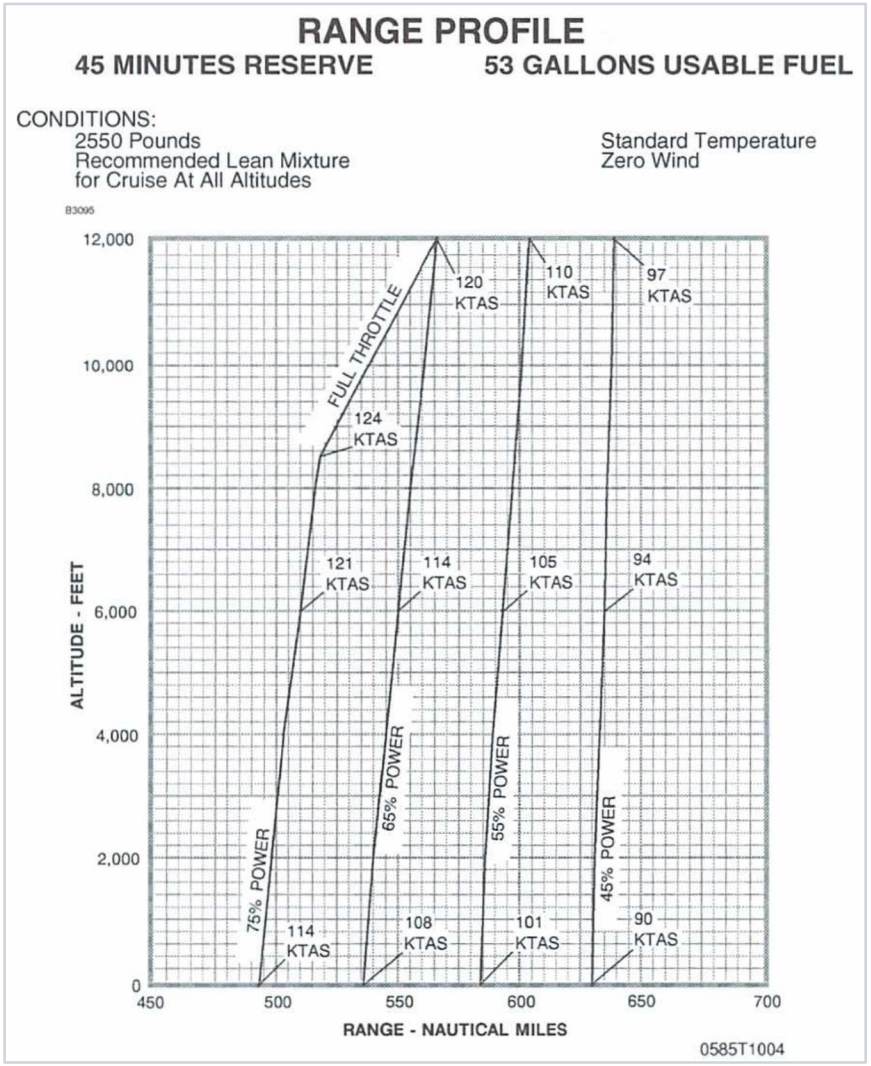
Cruise time: 342 nm divided by 107 knots = 3.2 hours. Cruise fuel: 3.2 hours times 8.9 GPH = 28.5 gallons (POH 5-7).

Reserve: 45 minutes divided by 60, times 8.9 GPH = 6.7 gallons (POH 5-7).

Total: 1.4 + 2.6 + 28.5 + 6.7 = 39.2 gallons, less than the 53 gallons usable (POH 5-7).

KNOW THIS FOR THE CHECKRIDE

The POH sample adds start, taxi, and takeoff fuel plus a 45 minute reserve (POH 5-7).



Source: POH 5-6, POH 5-7, POH Fig 5-9

Example: Landing Distance

AI.II.F.S1

- **Destination:** 2,000 ft, 25 C
- **Read at 2,000 ft, 30 C**
- **Ground roll:** 650 ft
- **Over 50 ft obstacle:** 1,455 ft

Destination: 2,000 feet pressure altitude, 25 degrees C, 3,000 foot field. The POH reads the chart at 2,000 feet and 30 degrees C: ground roll 650 feet, total over a 50 foot obstacle 1,455 feet (POH 5-8).

A wind correction can be made with Note 2 the same way as for takeoff (POH 5-8).

CONDITIONS:

Flaps Full
Power Off
Maximum Braking

Paved, level, dry runway
Zero Wind
Speed at 50 Ft: 61 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	545	1290	565	1320	585	1350	605	1380	625	1415
1000	565	1320	585	1350	605	1385	625	1420	650	1450
2000	585	1355	610	1385	630	1420	650	1455	670	1490
3000	610	1385	630	1425	655	1460	675	1495	695	1530
4000	630	1425	655	1460	675	1495	700	1535	725	1570
5000	655	1460	680	1500	705	1535	725	1575	750	1615
6000	680	1500	705	1540	730	1580	755	1620	780	1660
7000	705	1545	730	1585	760	1625	785	1665	810	1705
8000	735	1585	760	1630	790	1670	815	1715	840	1755

Source: POH 5-8, POH Fig 5-11

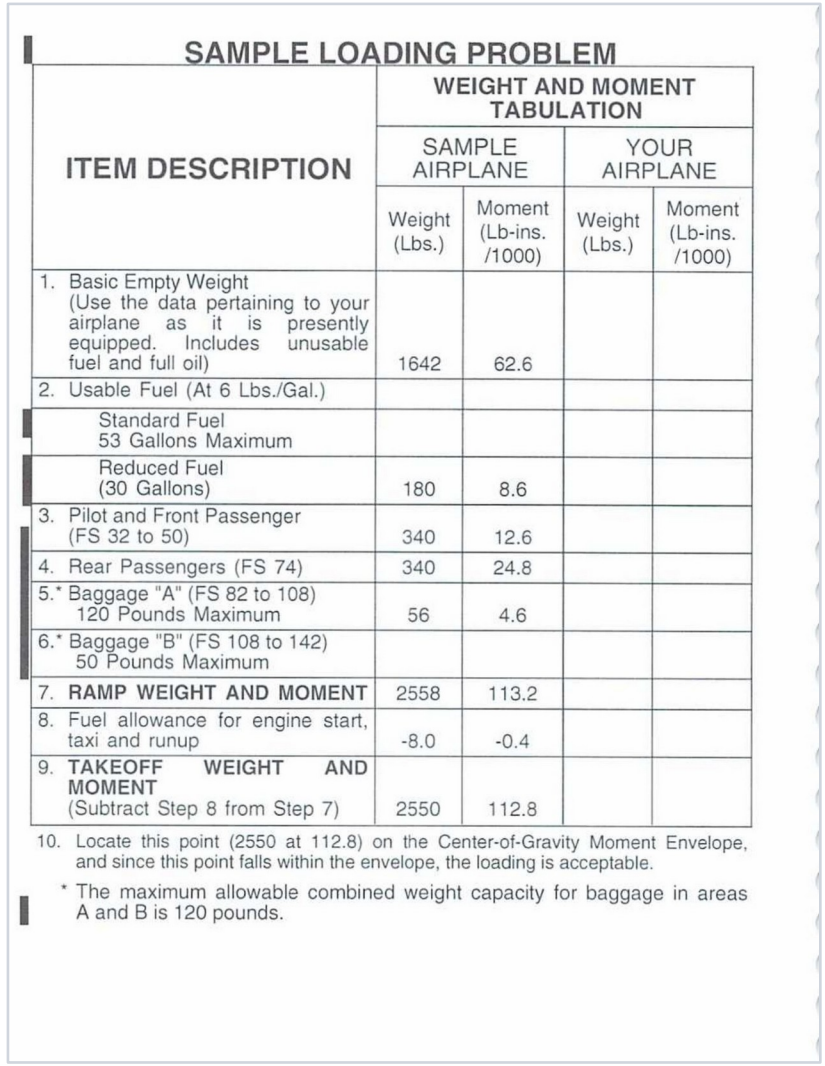
Example: C172S Sample Loading

AI.II.F.S2

Item	Weight (lb)	Moment /1000
Basic empty weight	1,642	62.6
Fuel (30 gal)	180	8.6
Pilot and front pass.	340	12.6
Rear passengers	340	24.8
Baggage A	56	4.6
Ramp	2,558	113.2
Start and taxi fuel	-8	-0.4
Takeoff	**2,550**	**112.8**

Walk through POH Figure 6-3 line by line. Basic empty weight includes unusable fuel and full oil. Fuel is figured at 6 lb per gallon (POH 6-10).

Ramp weight is 2,558 lb with a moment of 113.2. Subtract 8 lb and 0.4 moment for engine start, taxi, and runup. Takeoff weight is 2,550 lb with a moment of 112.8 (POH 6-10).



SAMPLE LOADING PROBLEM

ITEM DESCRIPTION	WEIGHT AND MOMENT TABULATION			
	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (Lbs.)	Moment (Lb-ins. /1000)	Weight (Lbs.)	Moment (Lb-ins. /1000)
1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil)	1642	62.6		
2. Usable Fuel (At 6 Lbs./Gal.)				
Standard Fuel 53 Gallons Maximum				
Reduced Fuel (30 Gallons)	180	8.6		
3. Pilot and Front Passenger (FS 32 to 50)	340	12.6		
4. Rear Passengers (FS 74)	340	24.8		
5.* Baggage "A" (FS 82 to 108) 120 Pounds Maximum	56	4.6		
6.* Baggage "B" (FS 108 to 142) 50 Pounds Maximum				
7. RAMP WEIGHT AND MOMENT	2558	113.2		
8. Fuel allowance for engine start, taxi and runup	-8.0	-0.4		
9. TAKEOFF WEIGHT AND MOMENT (Subtract Step 8 from Step 7)	2550	112.8		

10. Locate this point (2550 at 112.8) on the Center-of-Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

* The maximum allowable combined weight capacity for baggage in areas A and B is 120 pounds.

Source: POH 6-10, POH Fig 6-3

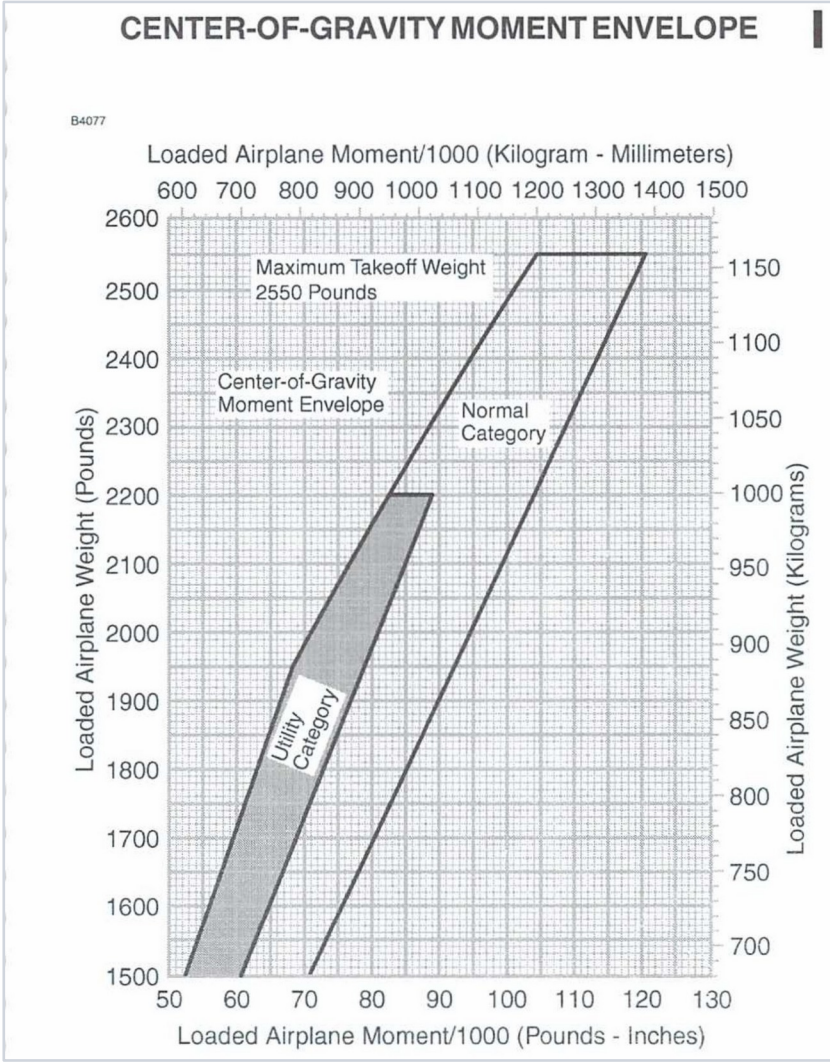
Example: Plot on the Envelope

AI.II.F.S2

- **Plot 2,550 lb at 112.8**
- **Inside envelope:** approved
- **CG:** 112,800 / 2,550 = 44.2 in
- **Limits:** 41.0 to 47.3 in

Locate 2,550 lb and 112.8 on the Center-of-Gravity Moment Envelope. The point falls within the envelope, so the loading is acceptable (POH 6-10, POH Fig 6-7).

To get the CG in inches, multiply the moment back by 1,000 and divide by weight: 112,800 divided by 2,550 = 44.2 inches. At 2,550 lb the limits are 41.0 to 47.3 inches (POH 2-8), so 44.2 is inside.



Source: POH 6-10, POH Fig 6-7, POH 2-8

Check Every Phase: Fuel Burn

AI.II.F.S2, AI.II.F.R6

- **Fuel arm: 48.0 in**
- **Burn 15 gal (90 lb)**

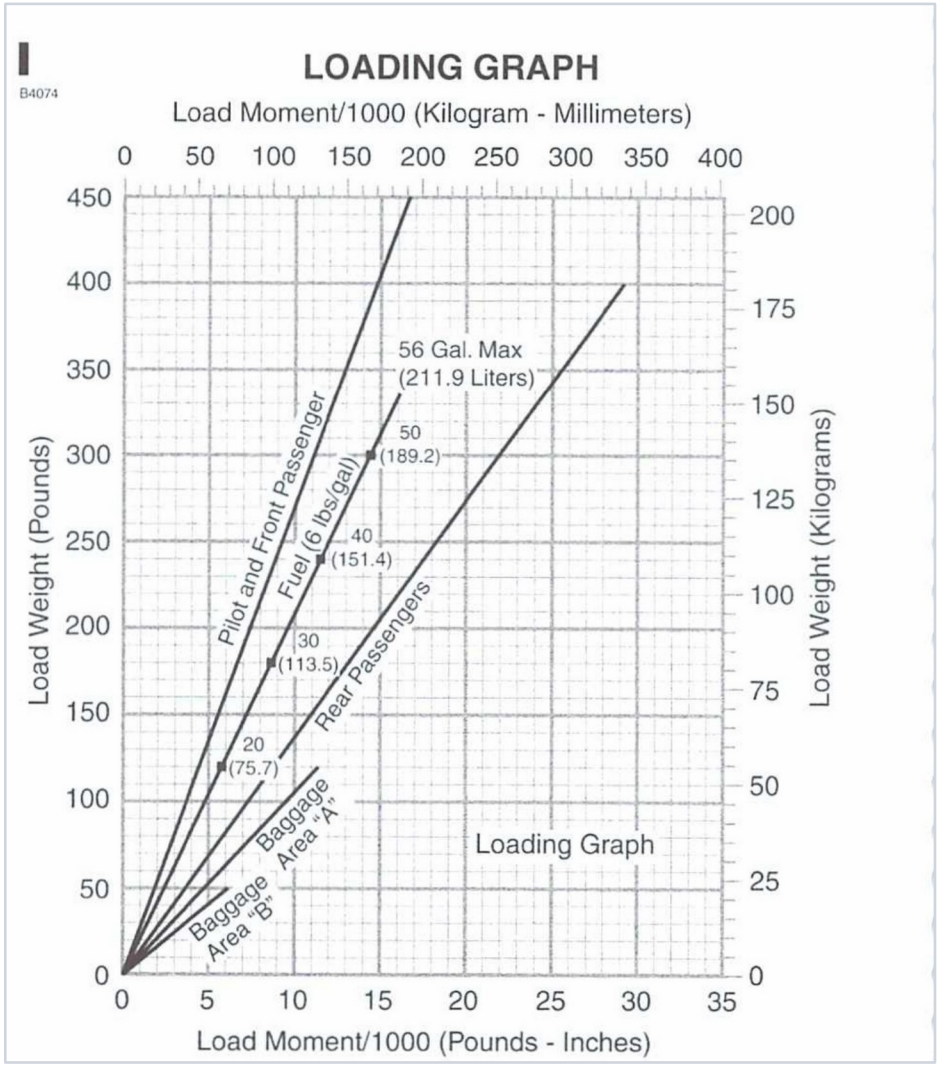
Phase	Weight	Moment /1000	CG (in)
Takeoff	2,550	112.8	44.2
Fuel burned	-90	-4.3	
Landing	**2,460**	**108.5**	**44.1**

The ACS requires checking weight and balance for all phases of flight. In flight, fuel burn is normally the only weight change (PHAK 10-2).

C172S fuel is in two wing tanks at 48.0 inches aft of datum (TCDS 3A12 p30). Most small aircraft have tanks close to the CG, so burning fuel does not move the CG much (PHAK 10-11).

EXAMPLE

Worked example using the POH sample load: burn 15 gallons (90 lb). Moment removed: 90 x 48.0 = 4,320 in-lb, or 4.3 on the /1000 scale. Landing: 2,460 lb and 108.5. CG: 108,500 divided by 2,460 = 44.1 inches. The forward limit at 2,460 lb is 35.0 plus (510 / 600 x 6.0) = 40.1 inches and the aft limit is 47.3 inches (POH 2-8), so the landing CG is in limits.



Source: TCDS 3A12 p30, PHAK 10-11, POH 2-8, POH Fig 6-4

Fixing an Out-of-Limits CG

AI.II.F.S2, AI.II.F.R6

- **Weight moved / total weight**
- **= CG change / distance moved**
- **PHAK:** move 65 lb forward
- **Recheck after every change**

Weight shift formula: weight shifted divided by total weight equals change of CG divided by distance the weight is shifted (PHAK 10-11).

PHAK example: total weight 7,800 lb, CG 81.5 inches, aft limit 80.5 inches. Cargo moves from station 150 to station 30, a 120 inch shift. Weight to be shifted / 7,800 = 1.0 / 120, so weight to be shifted = 65 lb (PHAK 10-11).

Adding or removing weight uses: added weight / new total weight = change of CG / distance between the weight and the old CG. If the CG moves aft, add the change. If it moves forward, subtract it (PHAK 10-11).

In the C172S, the practical fixes are moving bags forward, swapping heavier people to the front seats, or reducing weight (PHAK 10-3).

Example

Given:

Aircraft total weight . . . . . 7,800 lb
CG station. . . . . 81.5 in
Aft CG limit . . . . . 80.5 in

Determine how much cargo must be shifted from the aft cargo compartment at station 150 to the forward cargo compartment at station 30 to move the CG to exactly the aft limit.

Solution:

$$\frac{\text{Weight to be shifted}}{\text{Total weight}} = \frac{\Delta CG}{\text{Distance weight is shifted}}$$

$$\frac{\text{Weight to be shifted}}{7{,}800\ \text{lb}} = \frac{1.0\ \text{in}}{120\ \text{in}}$$

$$\text{Weight to be shifted} = 65\ \text{lb}$$

MEMORY AID

Weight moved / total weight = CG change / distance moved (PHAK 10-11).

Source: PHAK 10-10, PHAK 10-11

Quick Check

AI.II.F.S1, S2

1 **Climb fuel from 2,000 to 8,000 feet at 16 degrees above standard?**

Answer: 2.2 gallons plus 16 percent = 2.6 gallons (POH 5-6).

2 **Total fuel required for the POH sample trip?**

Answer: 39.2 gallons (POH 5-7).

3 **CG of the POH sample load at takeoff?**

Answer: 112,800 / 2,550 = 44.2 inches (POH 6-10).

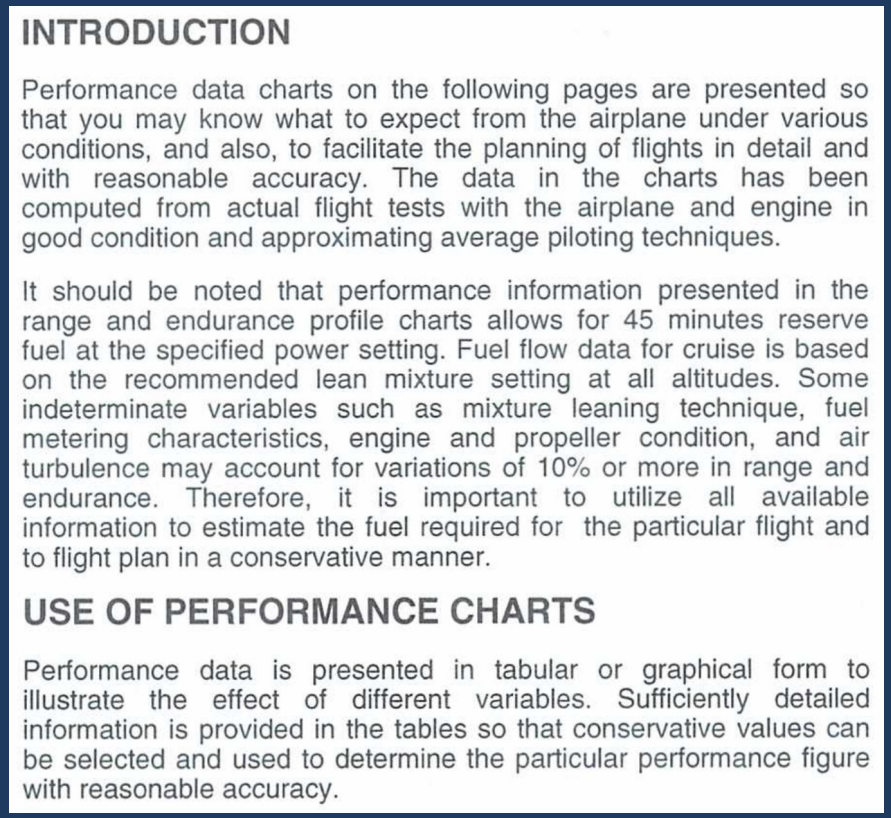
Source: POH 5-6, POH 5-7, POH 6-10

AI.II.F.R1 to R6

Risk Management

What you will learn

- Charts versus real life
- Limits and overweight
- CG limits and shifting weight



INTRODUCTION

Performance data charts on the following pages are presented so that you may know what to expect from the airplane under various conditions, and also, to facilitate the planning of flights in detail and with reasonable accuracy. The data in the charts has been computed from actual flight tests with the airplane and engine in good condition and approximating average piloting techniques.

It should be noted that performance information presented in the range and endurance profile charts allows for 45 minutes reserve fuel at the specified power setting. Fuel flow data for cruise is based on the recommended lean mixture setting at all altitudes. Some indeterminate variables such as mixture leaning technique, fuel metering characteristics, engine and propeller condition, and air turbulence may account for variations of 10% or more in range and endurance. Therefore, it is important to utilize all available information to estimate the fuel required for the particular flight and to flight plan in a conservative manner.

USE OF PERFORMANCE CHARTS

Performance data is presented in tabular or graphical form to illustrate the effect of different variables. Sufficiently detailed information is provided in the tables so that conservative values can be selected and used to determine the particular performance figure with reasonable accuracy.

Why it matters: the numbers are only as safe as the margins you add.

Source: POH 5-3, PHAK 10-1 to 10-4

Risk: Charts vs. Real Life

AI.II.F.R1, AI.II.F.R3

- **Expect 10% or more variation**
- **Use conservative chart values**
- **Set an abort point**
- **Scenario:** hot day, short strip

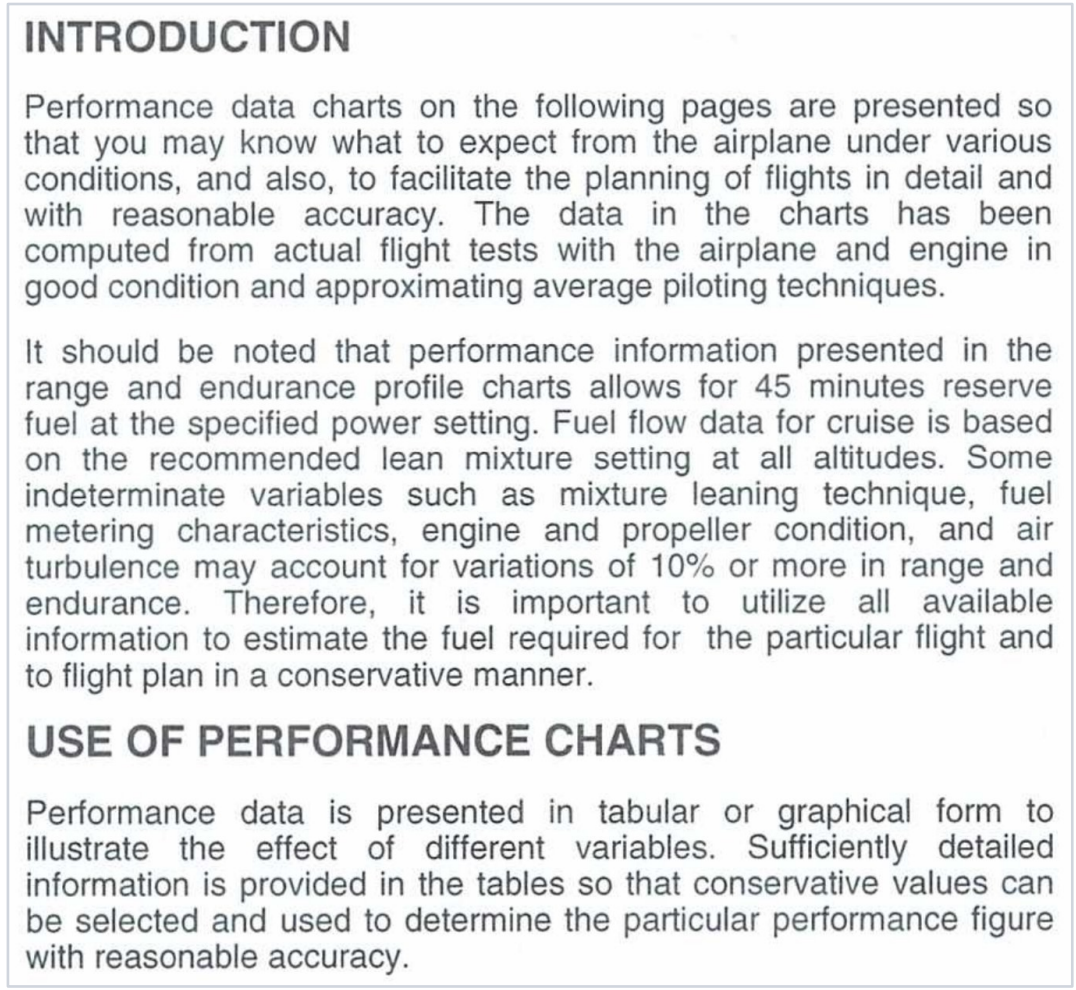
R1: Wrong chart, wrong row, or skipped notes give a wrong answer. Always read conditions and notes and use conservative values (POH 5-4).

R3: The POH says variables such as leaning technique, engine and propeller condition, and turbulence can cause variations of 10 percent or more in range and endurance, so plan conservatively (POH 5-3). Charts come from a new airplane flown with average skill (PHAK 11-19).

Mitigation: add a personal margin, plan an abort point on the runway, and decide before takeoff what you will do if you are not at the expected speed.

EXAMPLE

Scenario for discussion: 32 degrees C, field elevation 3,000 feet, 2,400 foot grass strip, airplane near maximum weight. Have the student find the chart number, apply the grass note, then decide go or no-go and what margin they want.



INTRODUCTION

Performance data charts on the following pages are presented so that you may know what to expect from the airplane under various conditions, and also, to facilitate the planning of flights in detail and with reasonable accuracy. The data in the charts has been computed from actual flight tests with the airplane and engine in good condition and approximating average piloting techniques.

It should be noted that performance information presented in the range and endurance profile charts allows for 45 minutes reserve fuel at the specified power setting. Fuel flow data for cruise is based on the recommended lean mixture setting at all altitudes. Some indeterminate variables such as mixture leaning technique, fuel metering characteristics, engine and propeller condition, and air turbulence may account for variations of 10% or more in range and endurance. Therefore, it is important to utilize all available information to estimate the fuel required for the particular flight and to flight plan in a conservative manner.

USE OF PERFORMANCE CHARTS

Performance data is presented in tabular or graphical form to illustrate the effect of different variables. Sufficiently detailed information is provided in the tables so that conservative values can be selected and used to determine the particular performance figure with reasonable accuracy.

Source: POH 5-3, POH 5-4, PHAK 11-19

Risk: Limits and Overweight

AI.II.F.R2, AI.II.F.R4

- **POH limits are mandatory**
- **Overweight:** poor climb, high stall
- **Structure not proven above max**
- **Scenario:** four adults, full tanks

R2: 14 CFR 91.9 requires the pilot in command to comply with the operating limits in the approved AFM, and these include weight and balance (PHAK 10-4).

Airspeed limits are limits too. Note the C172S maneuvering speed drops as weight drops: 105 KIAS at 2,550 lb, 98 KIAS at 2,200 lb, and 90 KIAS at 1,900 lb (POH 2-4).

R4: Operating above maximum weight compromises structural integrity and hurts performance (PHAK 10-1). An overloaded airplane may not leave the ground, or may show unusually poor flight characteristics (PHAK 10-2).

SYMBOL	SPEED	KCAS	KIAS	REMARKS
V_{NE}	Never Exceed Speed	160	163	Do not exceed this speed in any operation.
V_{NO}	Maximum Structural Cruising Speed	126	129	Do not exceed this speed except in smooth air, and then only with caution.
V_A	Maneuvering Speed: 2550 Pounds 2200 Pounds 1900 Pounds	 102 95 88	 105 98 90	Do not make full or abrupt control movements above this speed.
V_{FE}	Maximum Flap Extended Speed: Flaps 10° Flaps 10° to FULL	 107 85	 110 85	Do not exceed this speed with flaps down.
-----	Maximum Window Open Speed	160	163	Do not exceed this speed with windows open.

EXAMPLE

Scenario for discussion: four adults and bags for a weekend trip, full tanks. Have the student compute it with the POH sample empty weight, find the overage, and choose a fix (less fuel and a fuel stop, fewer bags, or fewer people).

Source: PHAK 10-1, PHAK 10-2, PHAK 10-4, POH 2-4

Risk: CG Limits and Shifting Weight

AI.II.F.R5, AI.II.F.R6

- **Out of CG:** control trouble
- **Aft CG:** stall and spin danger
- **Moved bags move the CG**
- **Scenario:** heavy bags in back

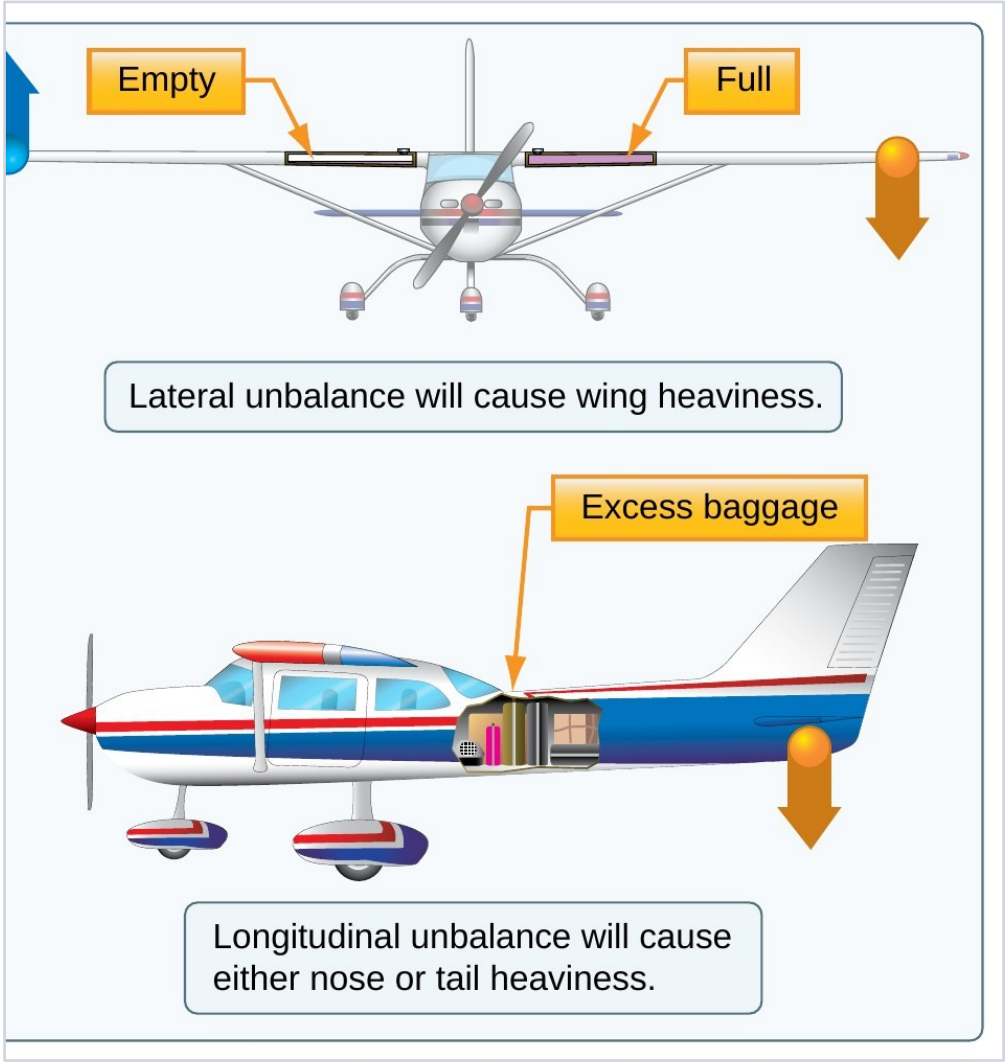
R5: Operation with the CG outside the approved limits results in control difficulty (PHAK 10-1). Aft CG can cause violent stall characteristics and very light control forces (PHAK 10-3).

R6: The CG depends on how weight is distributed. As load is shifted or used, the CG moves (PHAK 10-3). Adding or removing weight changes both weight and CG (PHAK 10-11).

Baggage area limits: area A holds 120 lb and area B holds 50 lb, with 120 lb combined maximum (POH 2-8). Exceeding these overloads the floor even if the total weight is fine.

EXAMPLE

Scenario for discussion: after you finish the weight and balance, a passenger adds a 40 lb cooler to baggage area B. Ask what changed, whether the area B limit is exceeded with what is already there, and how to recompute.



Source: PHAK 10-1, PHAK 10-3, PHAK 10-11, POH 2-8, PHAK Fig 10-1

Quick Check

AI.II.F.R1 to R6

1 **How much can real performance differ from the charts?**

Answer: The POH says variations of 10 percent or more in range and endurance (POH 5-3).

2 **Which regulation requires following the AFM limits?**

Answer: 14 CFR 91.9 (PHAK 10-4).

3 **A passenger adds a bag after you finished the weight and balance. What now?**

Answer: Recompute. Adding weight changes both weight and CG (PHAK 10-11).

Source: POH 5-3, PHAK 10-4, PHAK 10-11

Common Errors

AI.II.F.S1, AI.II.F.S2

- **Using the sample empty weight**
- **Skipping the chart notes**
- **No temperature correction**
- **Forgetting taxi fuel**

Using the POH sample empty weight. Fix: use the airplane's current weight and balance record. Sample problems are guidance only (PHAK 10-6).

Skipping the notes (wind, grass, flaps up). Fix: read every note before using a number (POH Fig 5-5).

Not correcting climb numbers for temperature. Fix: add 10 percent per 10 degrees C above standard (POH 5-6).

Forgetting start and taxi fuel. Fix: subtract 8 lb at ramp for weight and balance (POH 6-10) and add 1.4 gallons for fuel planning (POH Fig 5-7).

TIME, FUEL AND DISTANCE TO CLIMB AT 2550 POUNDS

CONDITIONS:

Flaps Up
Full Throttle
Standard Temperature

PRESS ALT FT	CLIMB SPEED KIAS	RATE OF CLIMB FPM	FROM SEA LEVEL		
			TIME IN MIN	FUEL USED GAL	DIST NM
S.L.	74	730	0	0.0	0
1000	73	695	1	0.4	2
2000	73	655	3	0.8	4
3000	73	620	4	1.2	6
4000	73	600	6	1.5	8
5000	73	550	8	1.9	10
6000	73	505	10	2.2	13
7000	73	455	12	2.6	16
8000	72	410	14	3.0	19
9000	72	360	17	3.4	22
10,000	72	315	20	3.9	27
11,000	72	265	24	4.4	32
12,000	72	220	28	5.0	38

NOTE

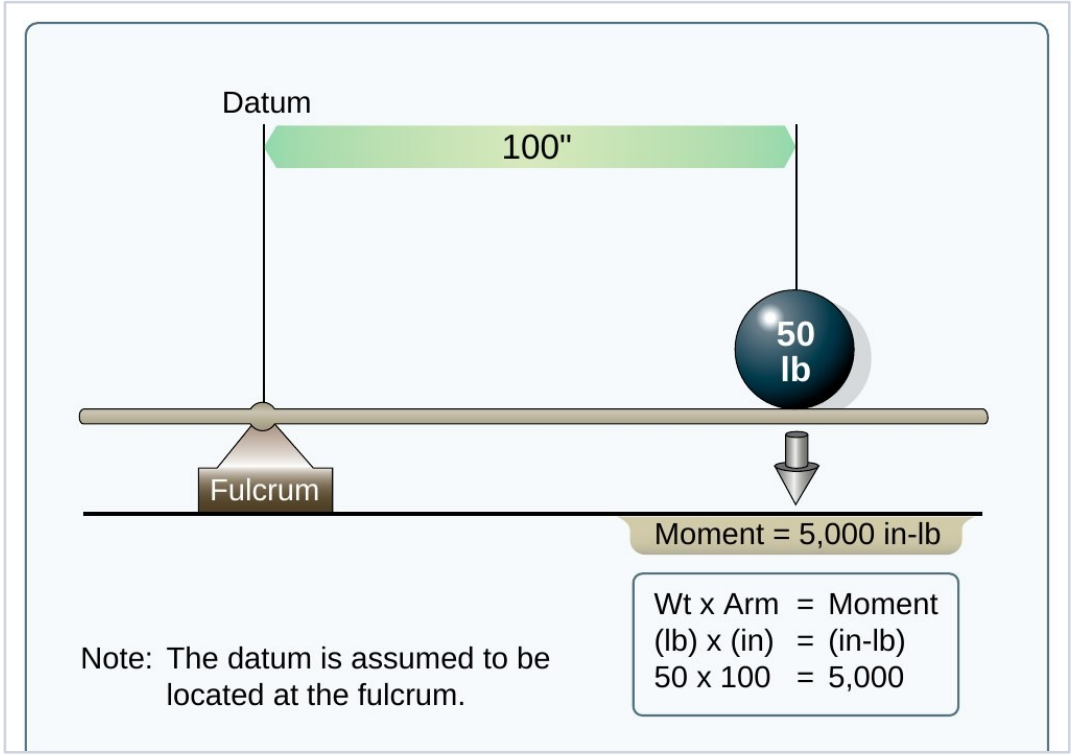
- Add 1.4 gallons of fuel for engine start, taxi and takeoff allowance.
- Mixture leaned above 3000 feet elevation for maximum RPM.
- Increase time, fuel and distance by 10% for each 10°C above standard temperature.
- Distances shown are based on zero wind.

Source: PHAK 10-6, POH 5-4, POH 5-6, POH 6-10, POH Fig 5-5, POH Fig 5-7

Review Questions

AI.II.F

- **What raises density altitude?**
- **Where is the C172S datum?**
- **How do you find CG?**
- **18 kt headwind:** what change?

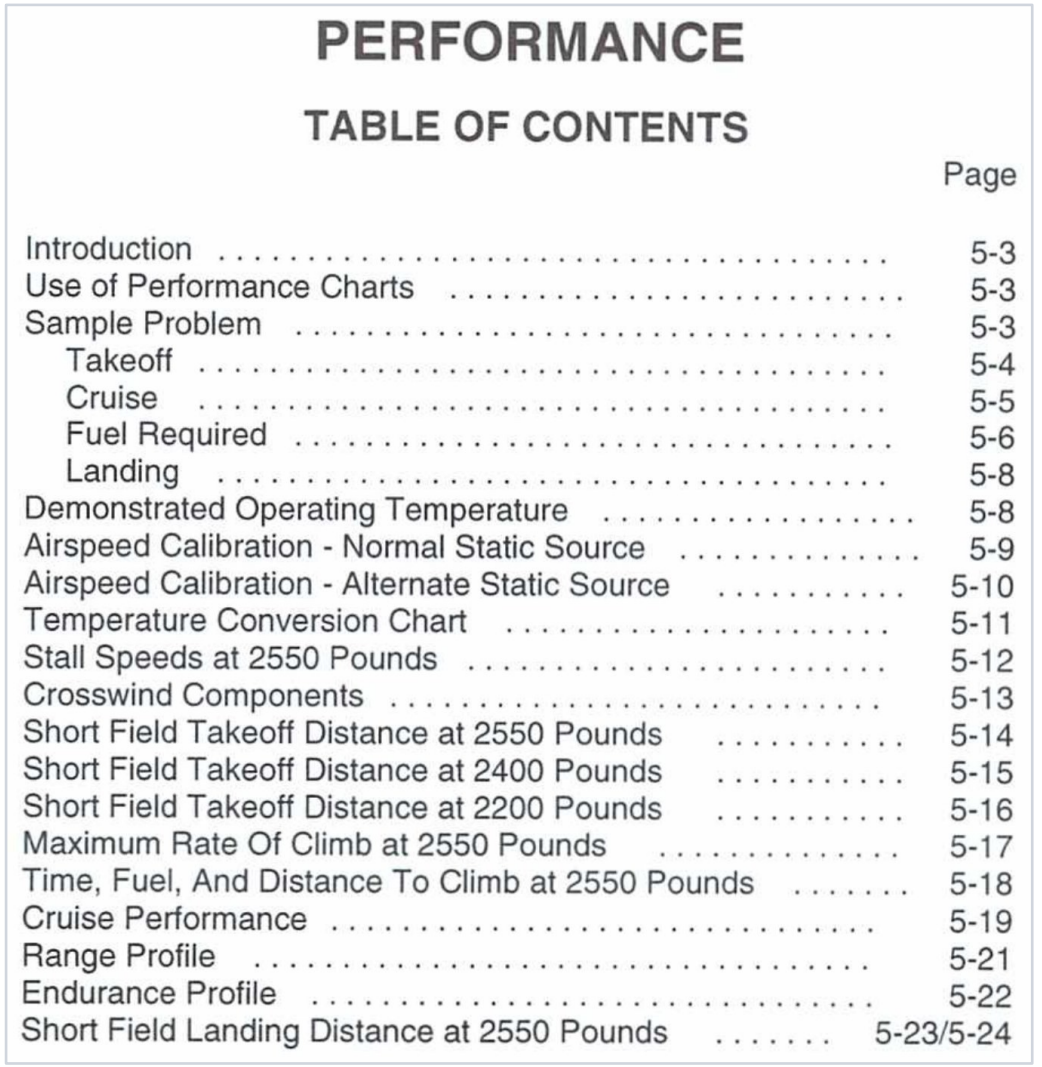


Source: PHAK 11-2, PHAK 10-7, TCDS 3A12 p30, POH Fig 5-5

References

AI.II.F

- FAA-S-ACS-25, Task II.F
- PHAK Ch 5, 10, 11
- C172S POH Sec 2, 5, 6
- TCDS 3A12 Rev 89



PERFORMANCE

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Source: See lesson plan for full reference list